

ACTIVE DIRECTORY DELOPMENT

Fig 1: Open the Server Manager

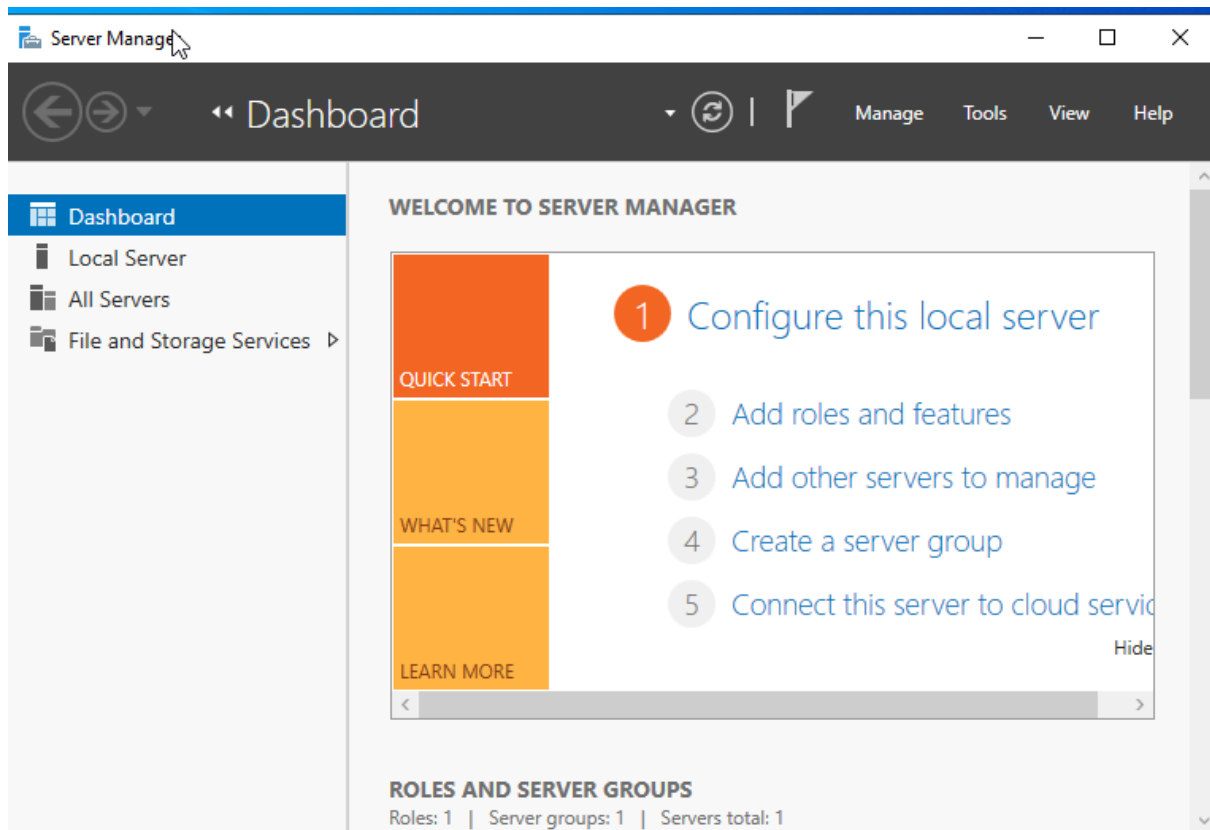


Fig 2: Navigate to Manage tab and select **Add Roles and Features**

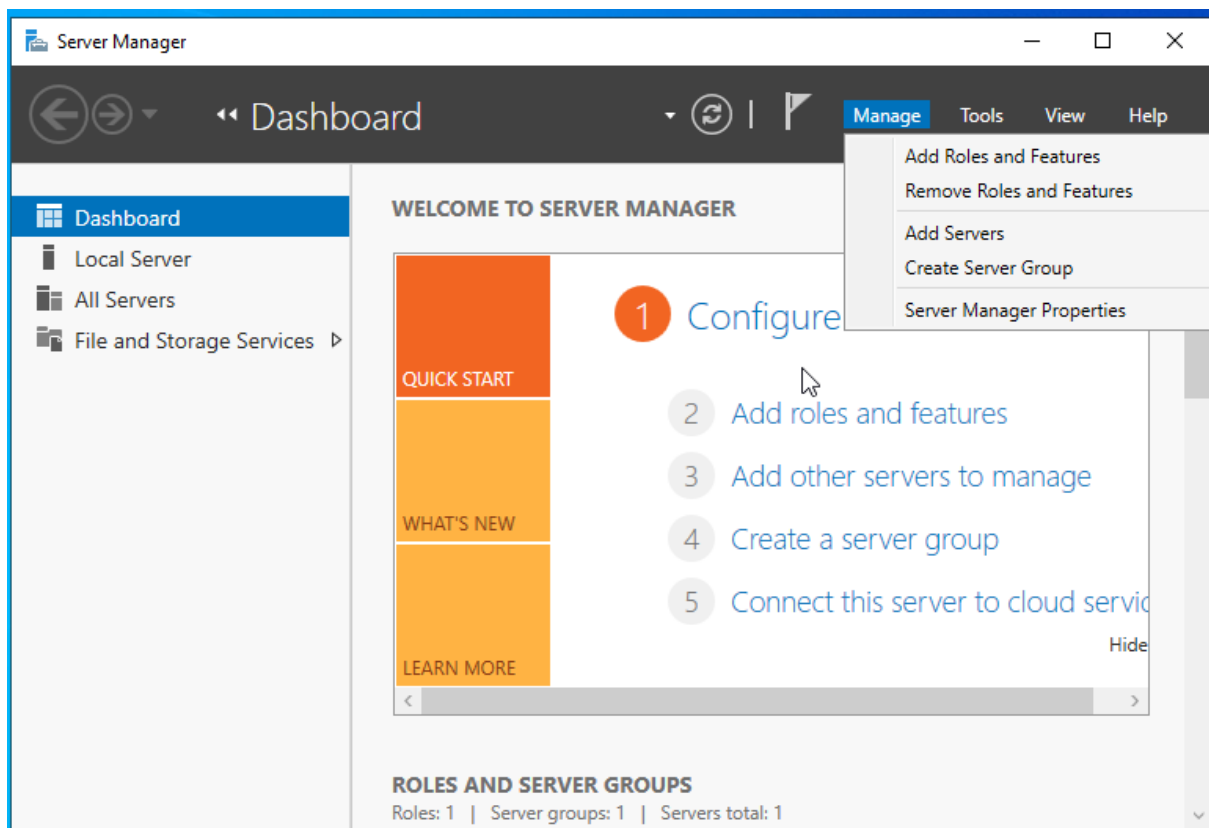


Fig 3: Click Next

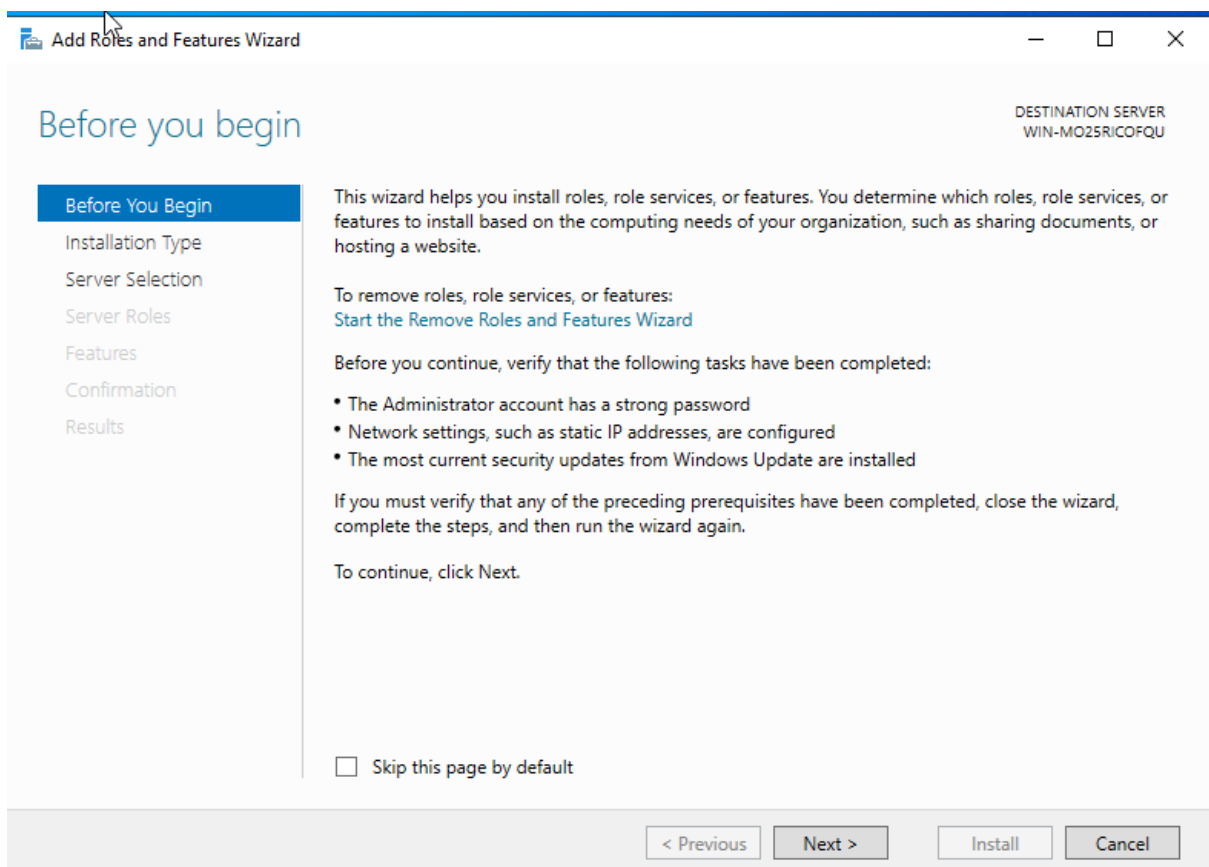


Fig 4: Click Next

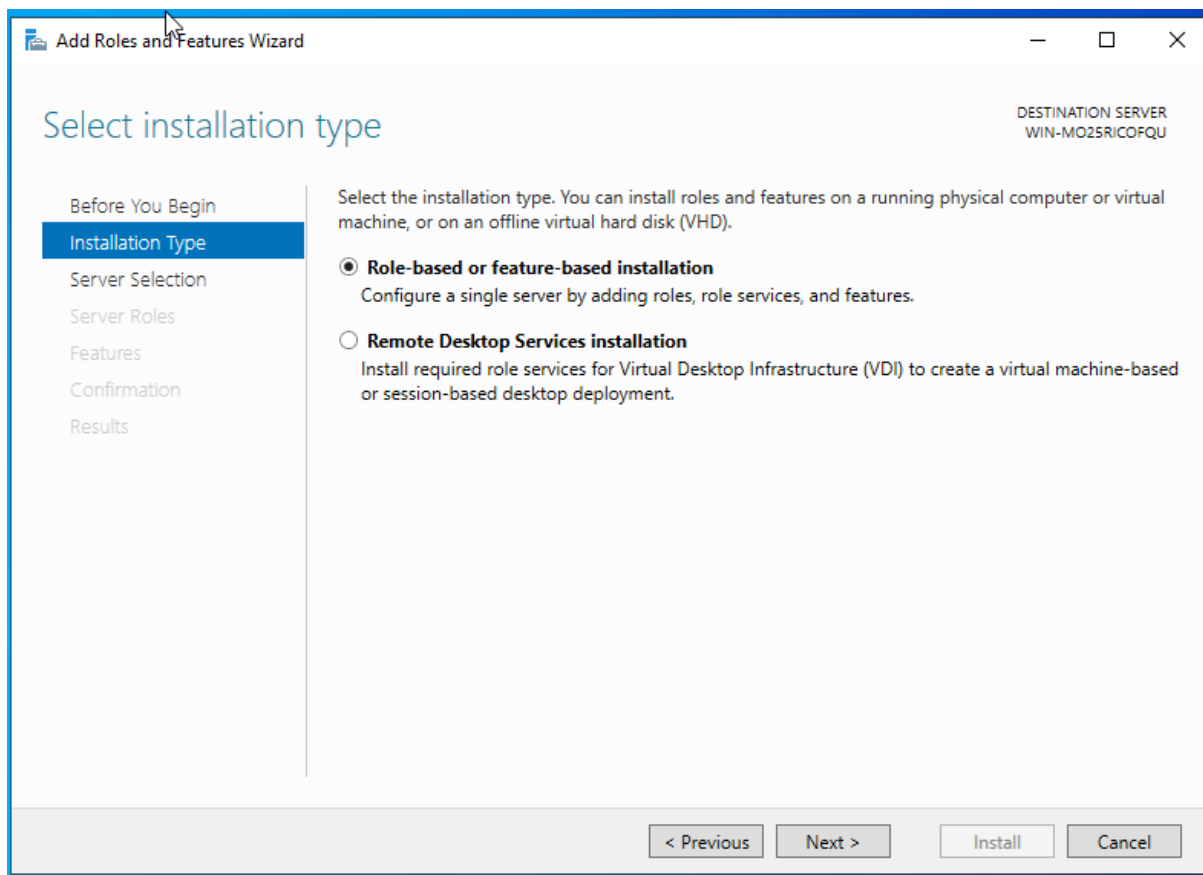


Fig 5: Click Next

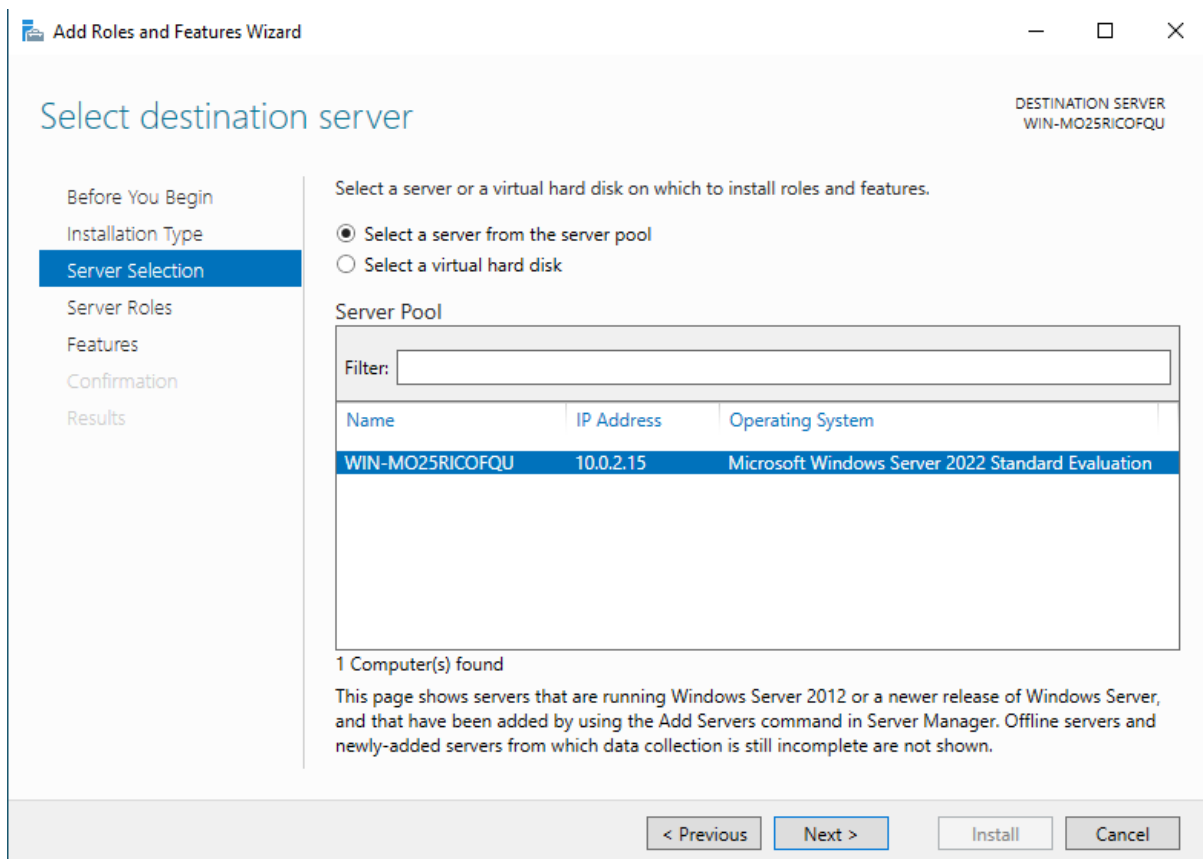


Fig 6: Click on the Active Directory Domain Service

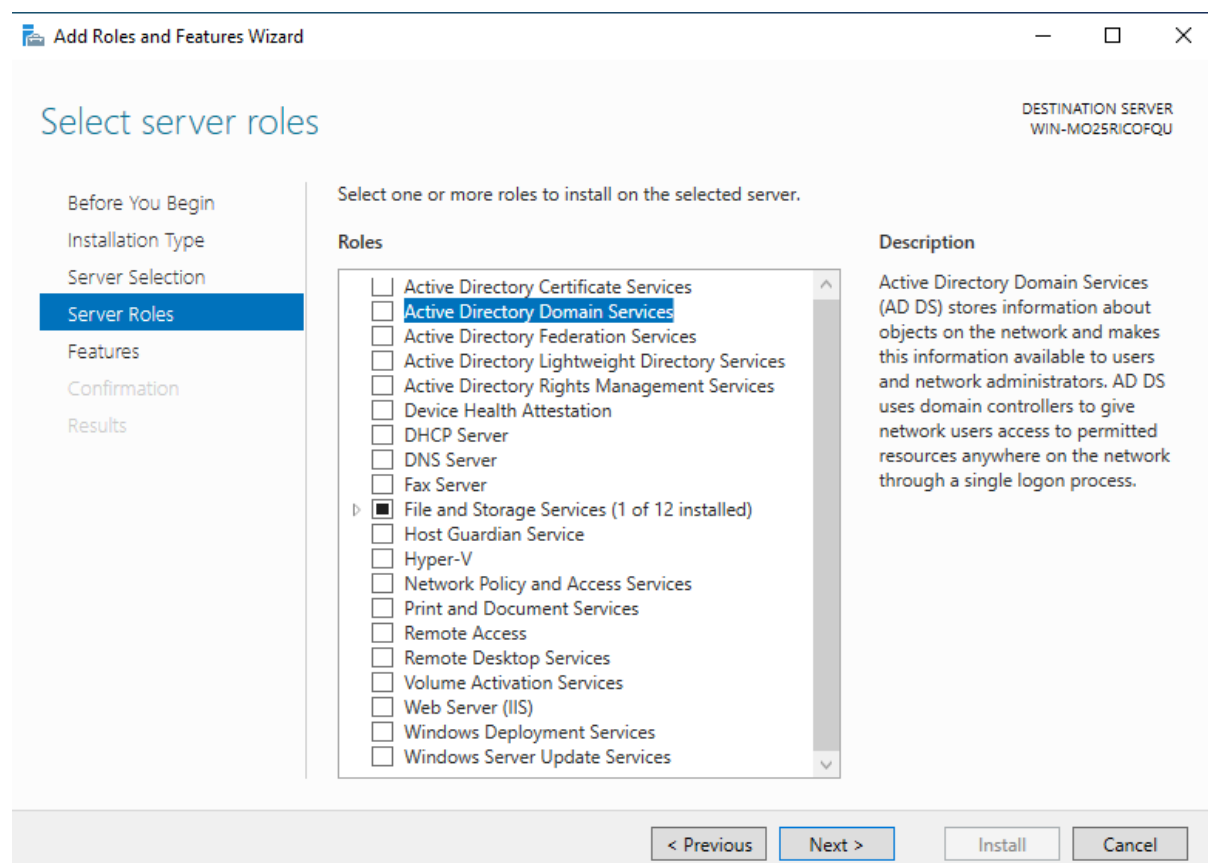


Fig 7: Click Add Features.

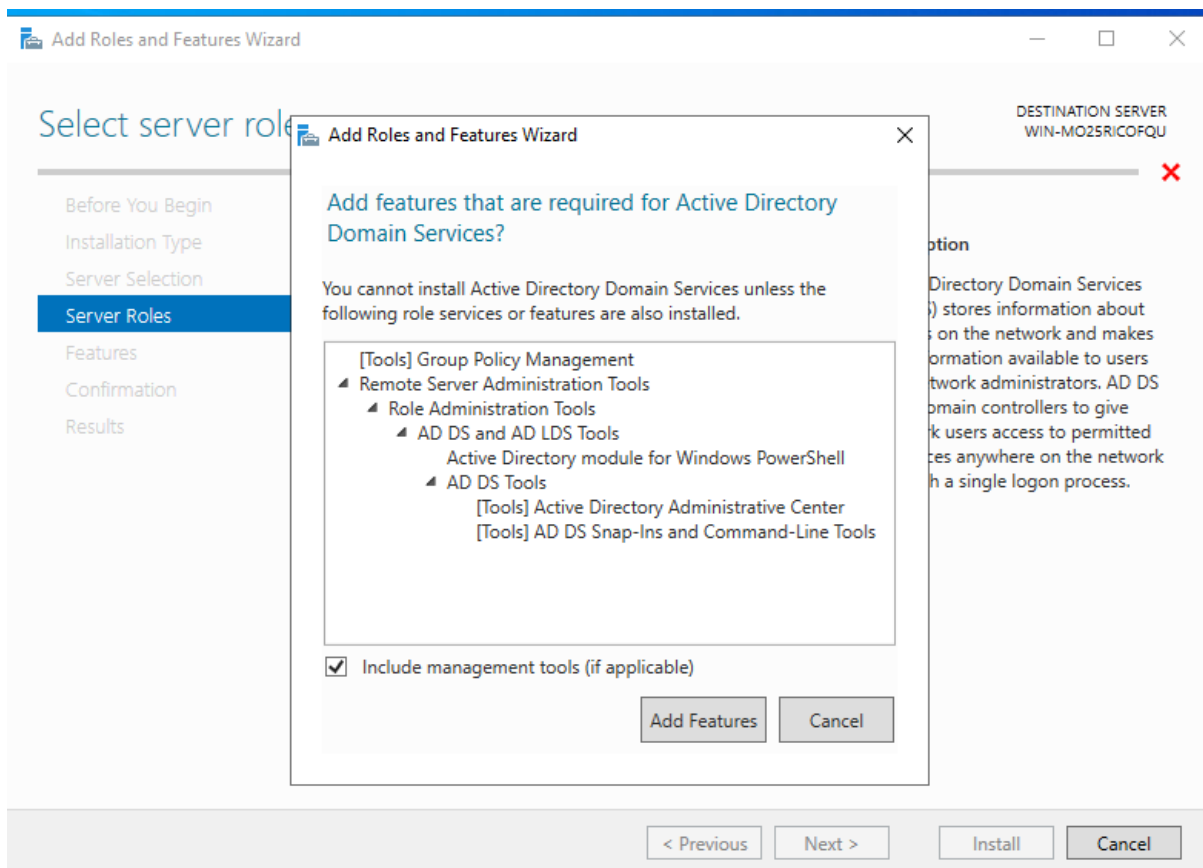


Fig 8: Click Next

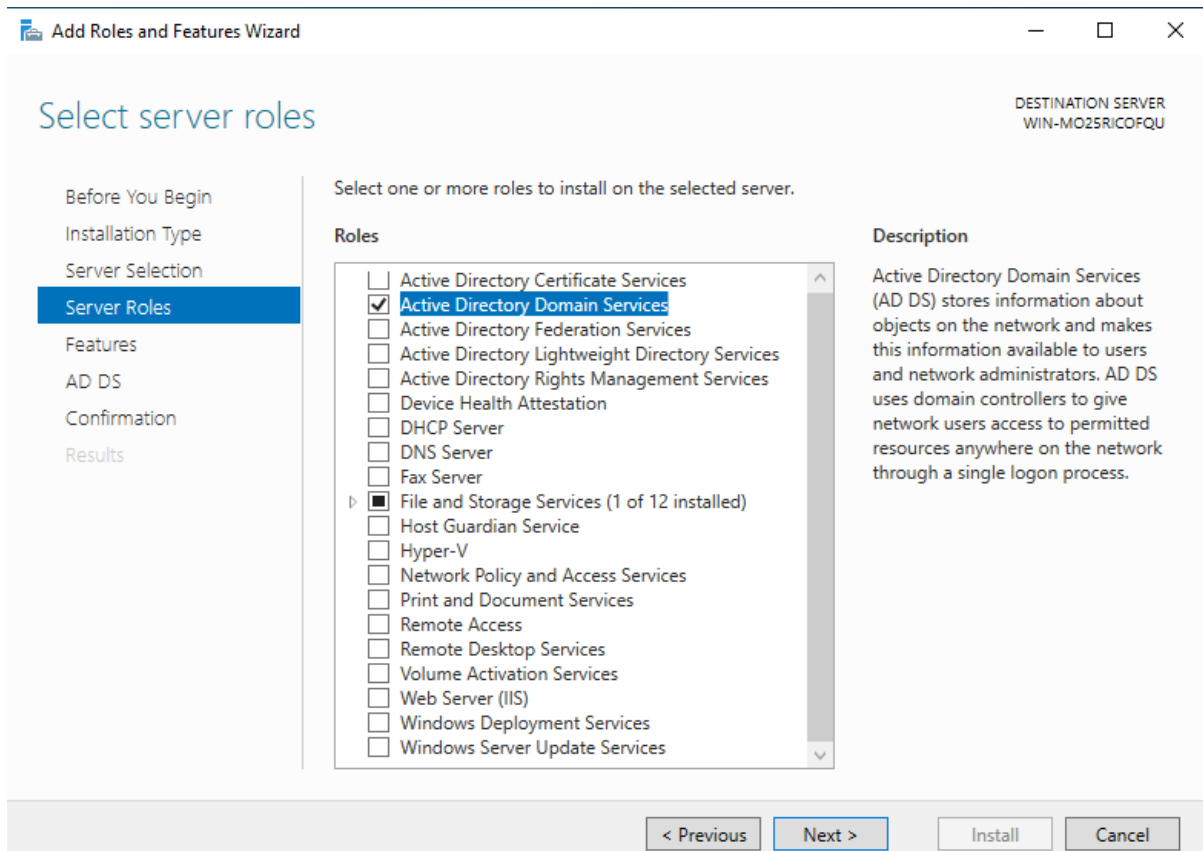


Fig 9: Click Next

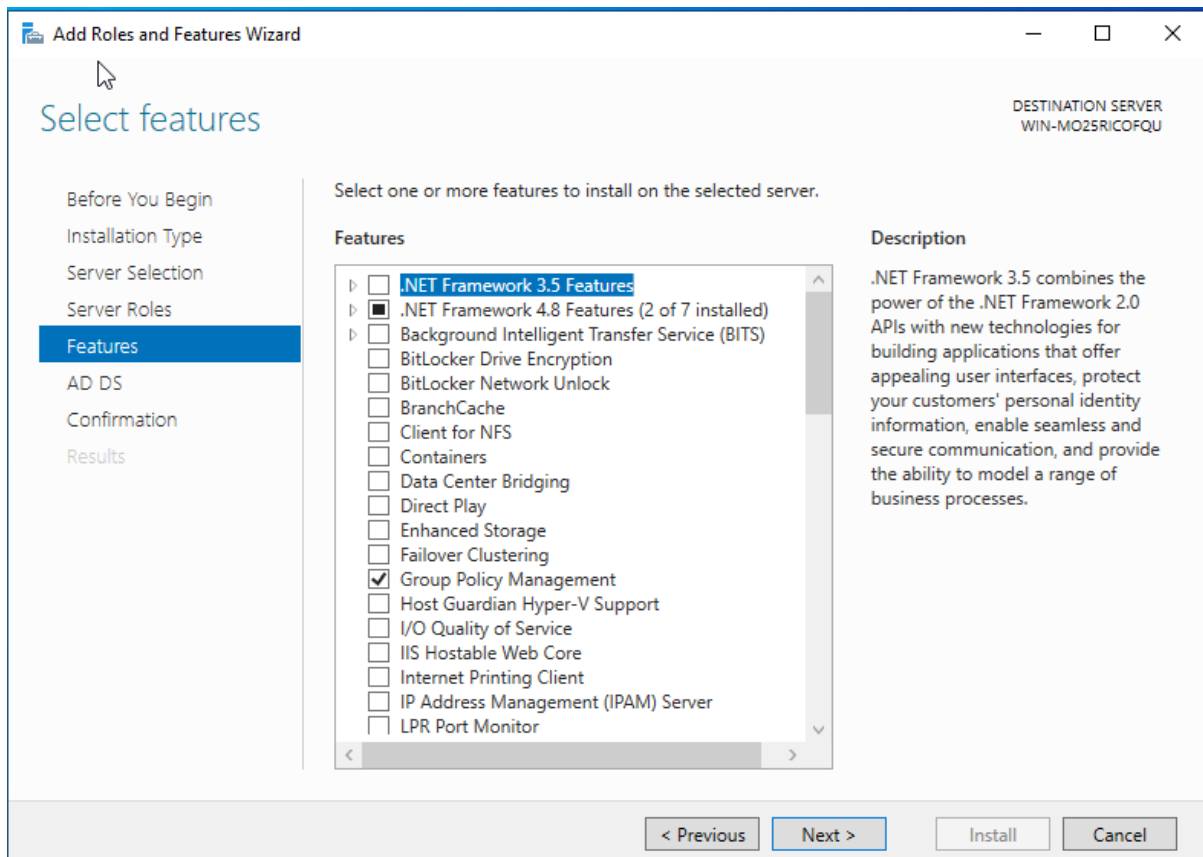


Fig 10: Click Next

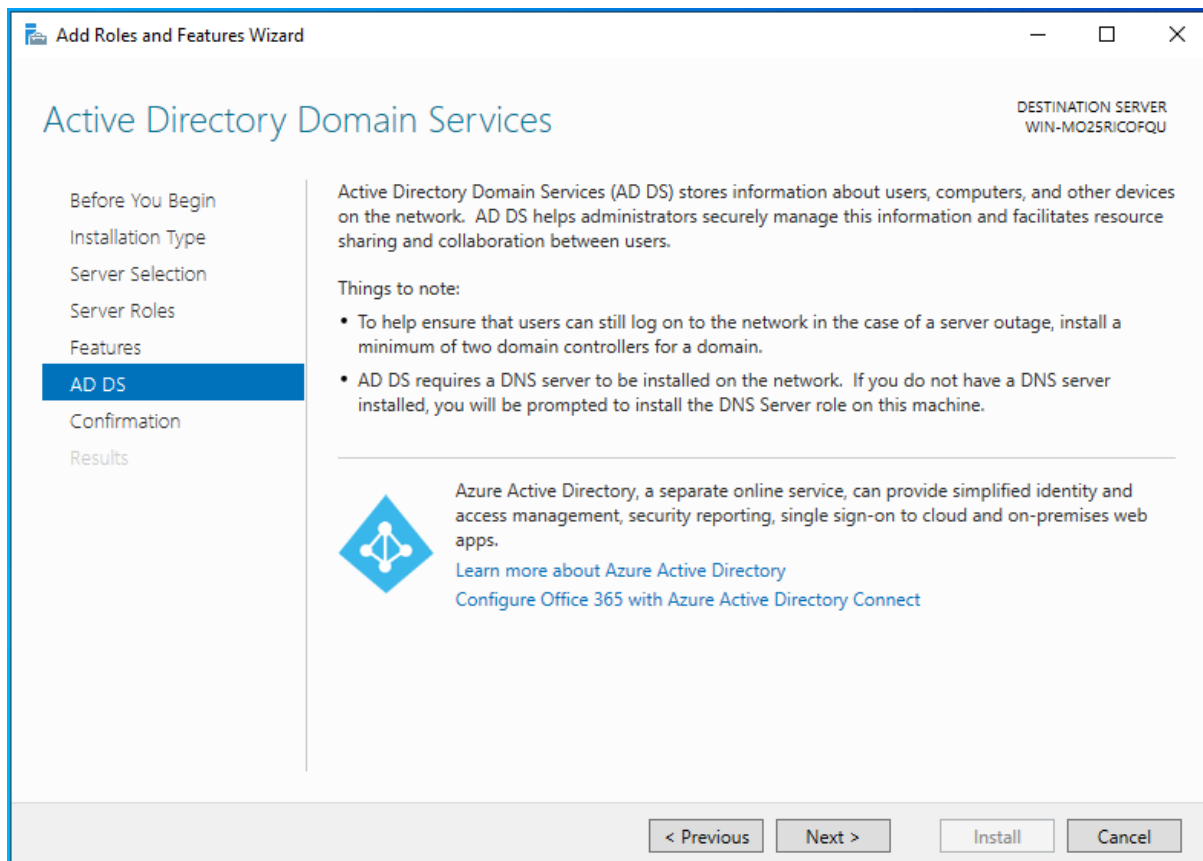


Fig 11: Click on Install

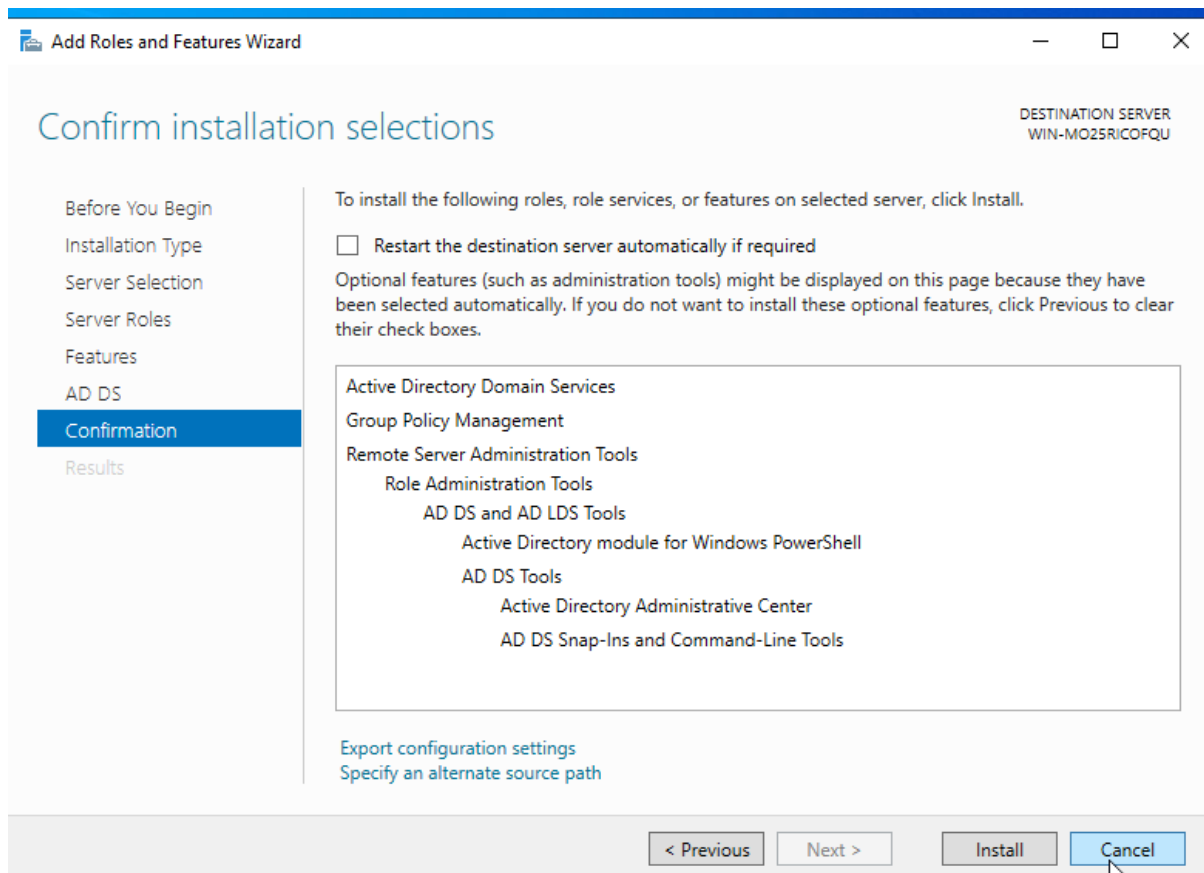


Fig 12: Installation in progress

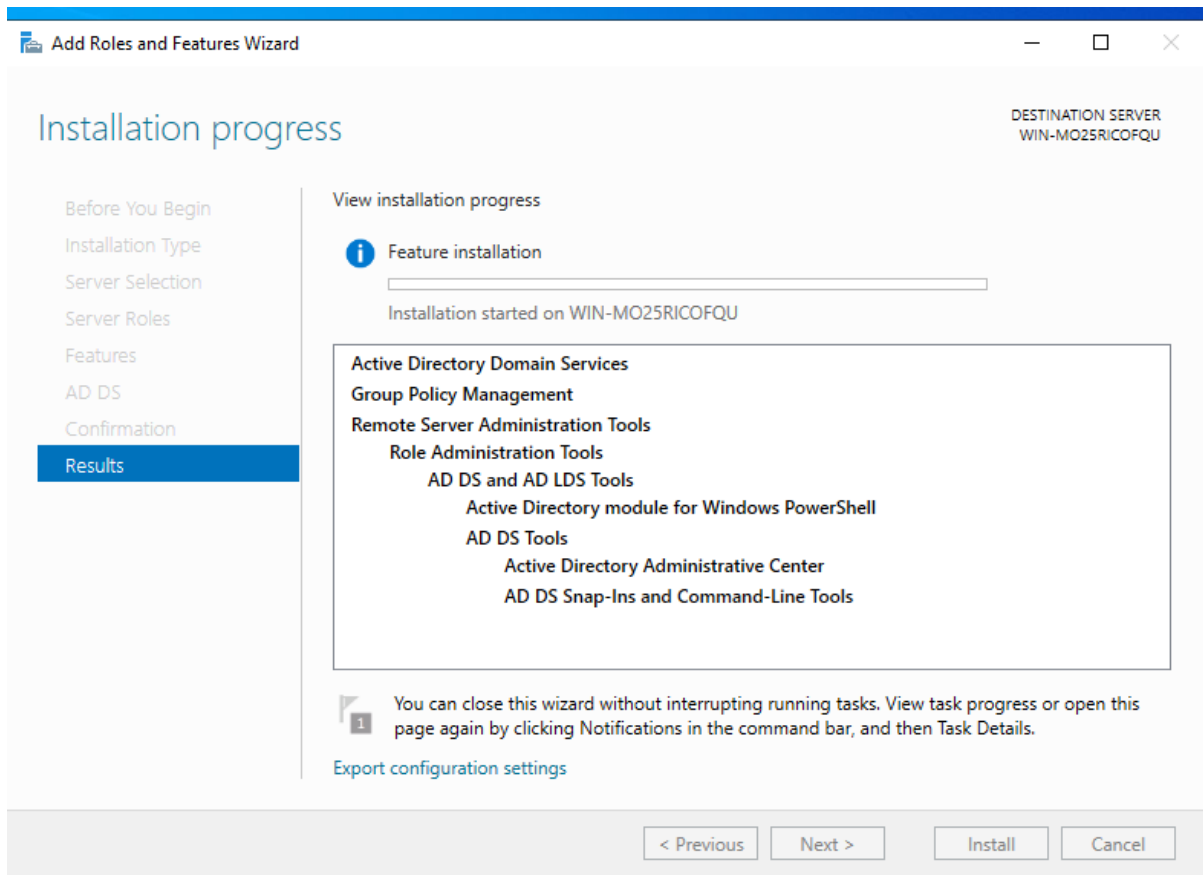


Fig 13: Installation completed

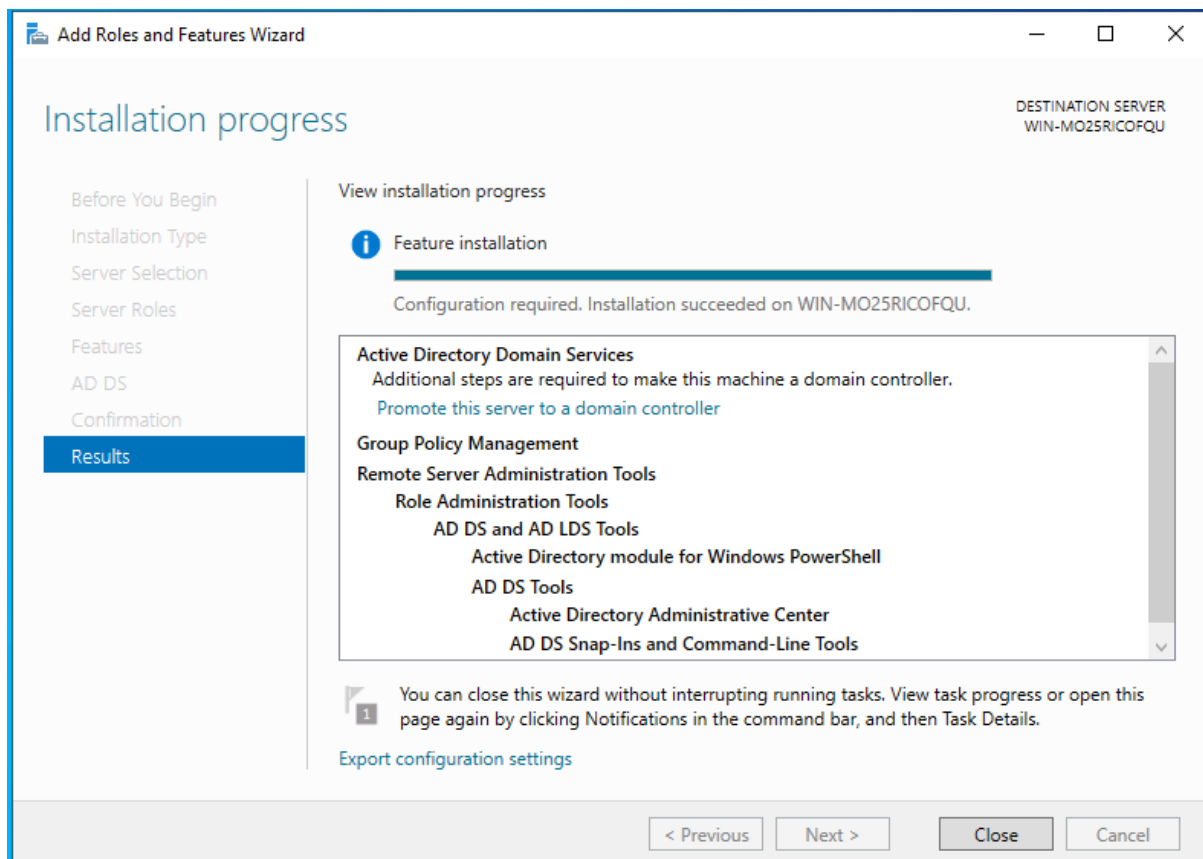


Fig 14: Click the notification icon by the Flag.

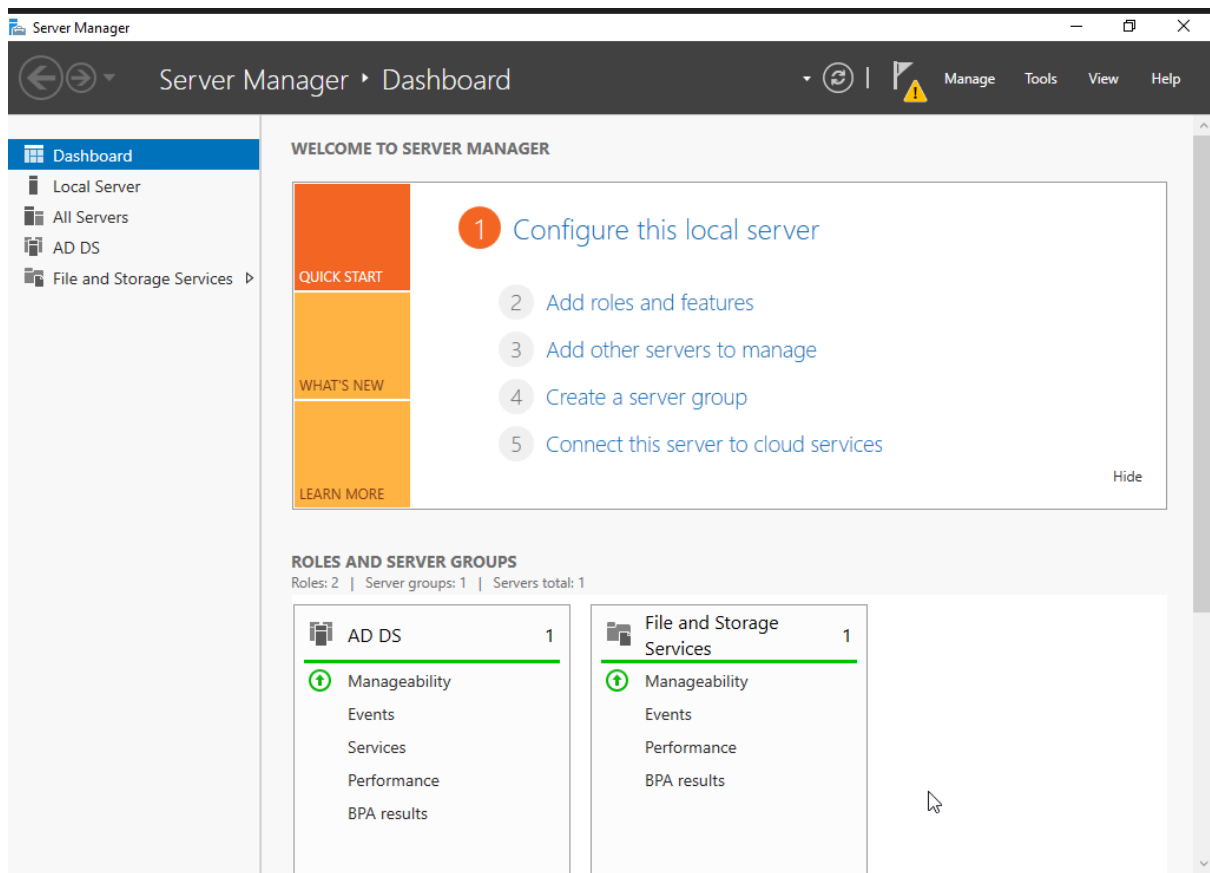


Fig 15: Post -deployment Configuration pop up displayed and click Promote this server to a domain controller.

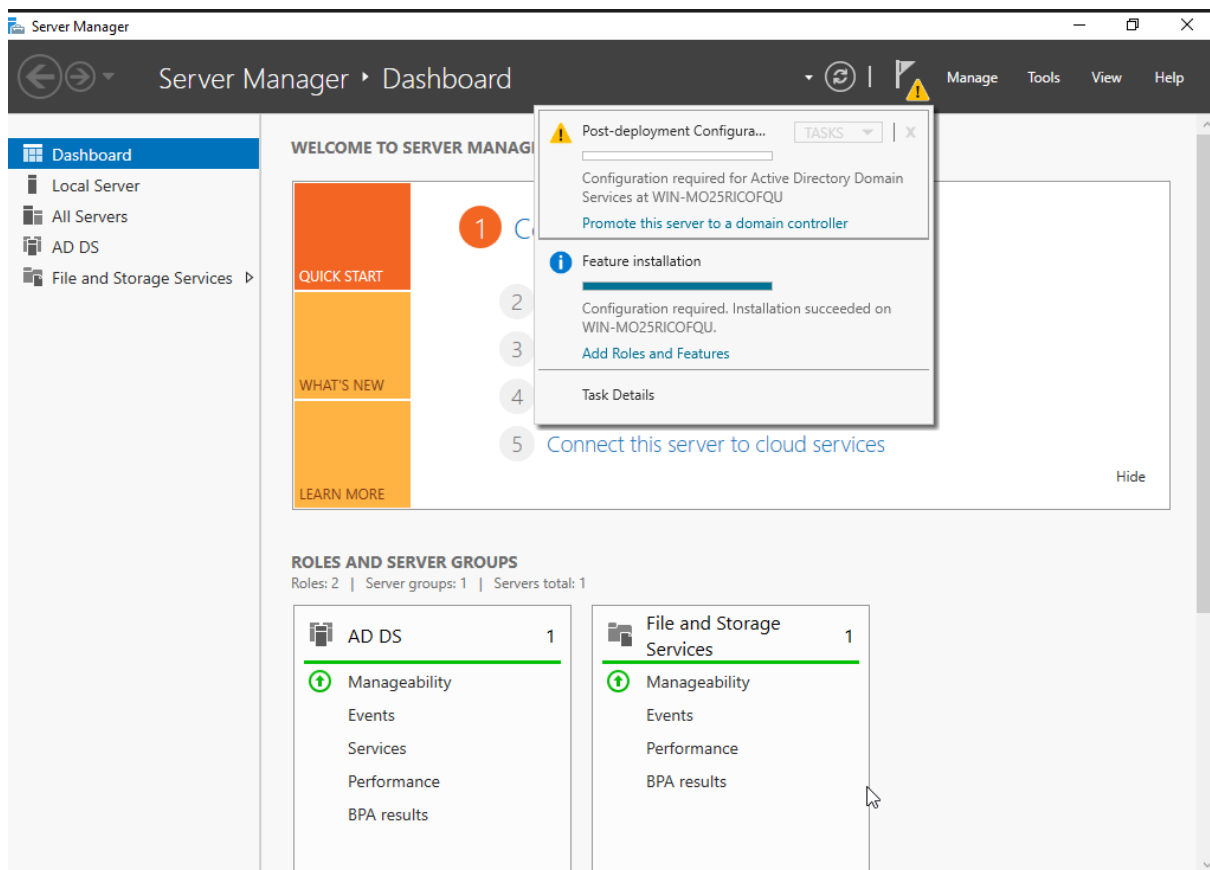


Fig 16: Add a new forest, enter a Root domain name, and click Next.

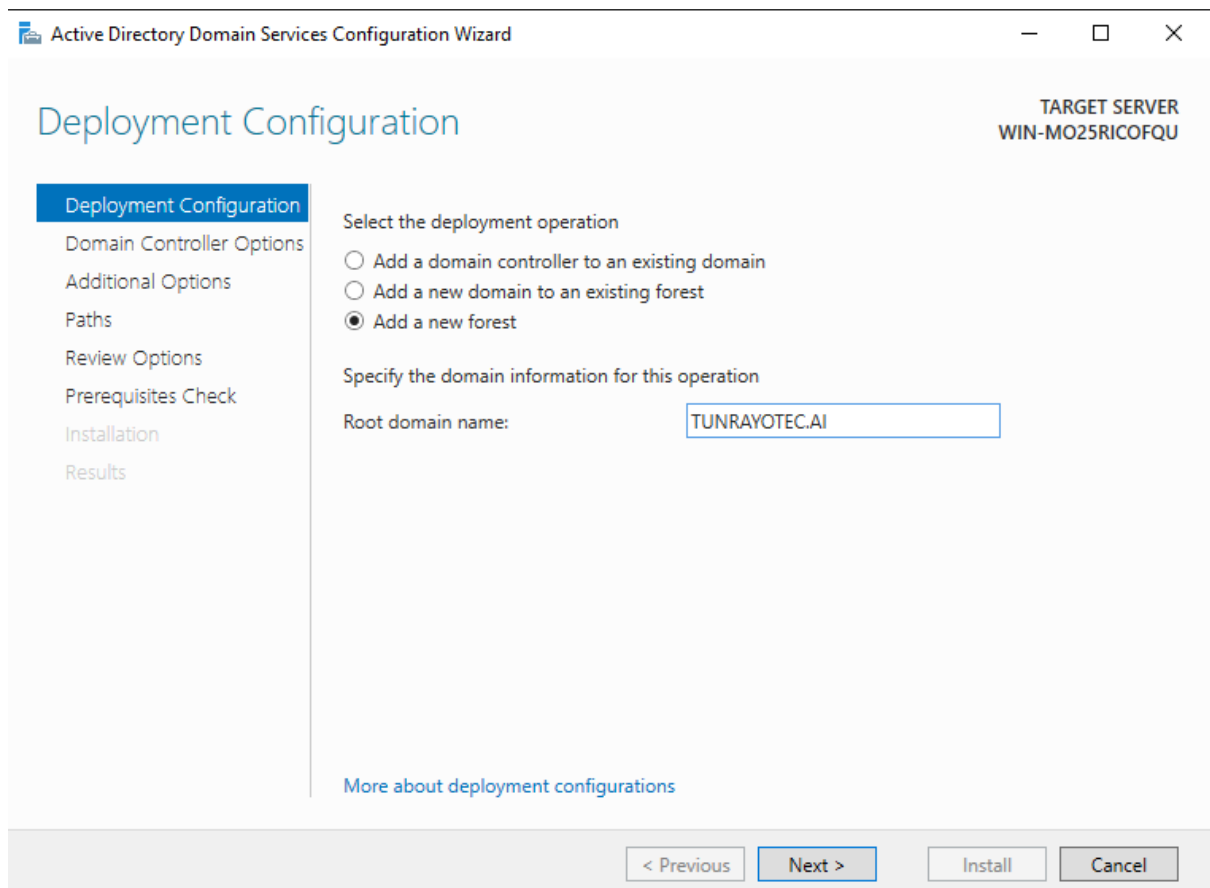


Fig 17: Enter password and click Next.

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the application name and standard window controls. The main window has a left-hand navigation pane with the following items: 'Deployment Configuration', 'Domain Controller Options' (highlighted in blue), 'DNS Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main content area is titled 'Domain Controller Options' and shows 'TARGET SERVER WIN-MO25RICOFQU'. It contains the following sections: 'Select functional level of the new forest and root domain' with 'Forest functional level' and 'Domain functional level' both set to 'Windows Server 2016'; 'Specify domain controller capabilities' with checkboxes for 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked); and 'Type the Directory Services Restore Mode (DSRM) password' with 'Password:' and 'Confirm password:' fields, each with a red asterisk indicating a required field. At the bottom right of the main content area is a link 'More about domain controller options'. The bottom of the window features a navigation bar with four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
WIN-MO25RICOFQU

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server
☒ Global Catalog (GC)
☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password: *

Confirm password: *

[More about domain controller options](#)

< Previous Next > Install Cancel

Fig 18: Click Next

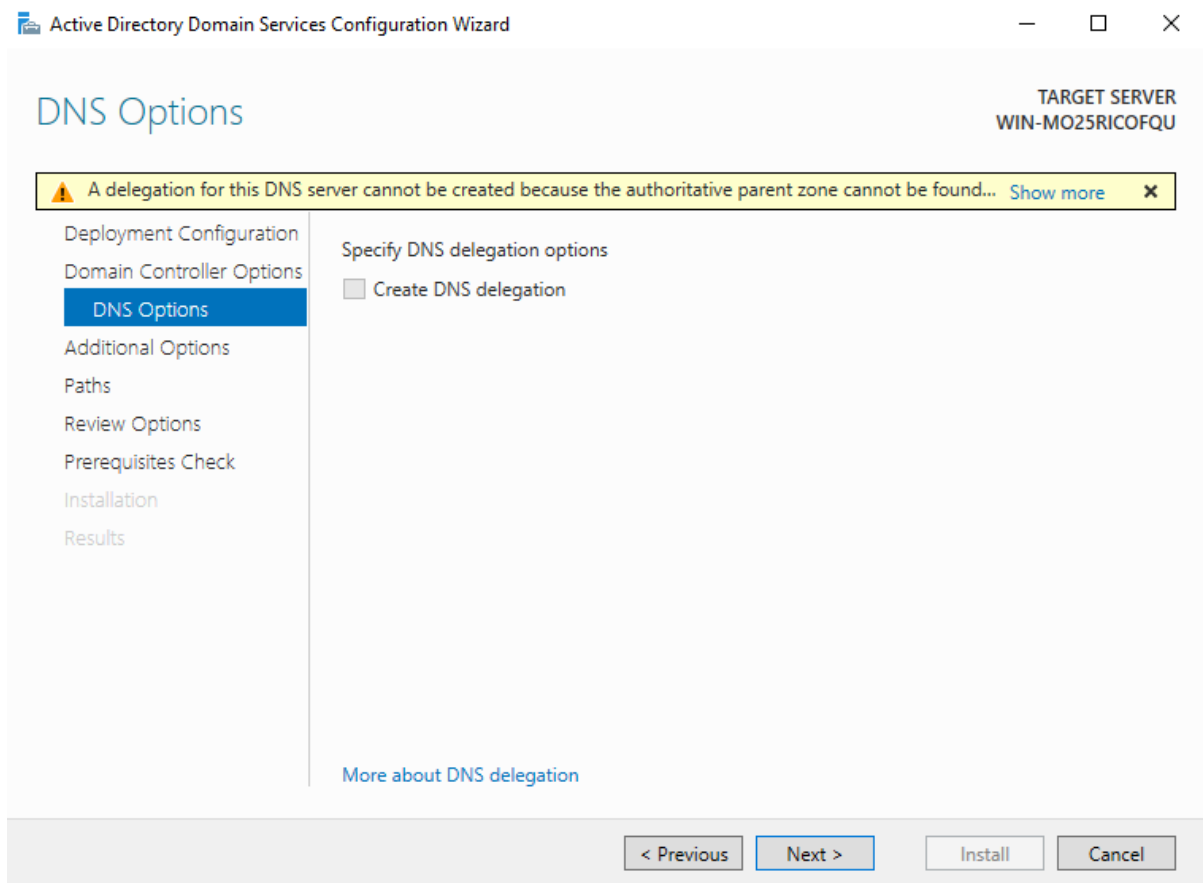


Fig 19: Click Next and by default the Domain Name is populated.

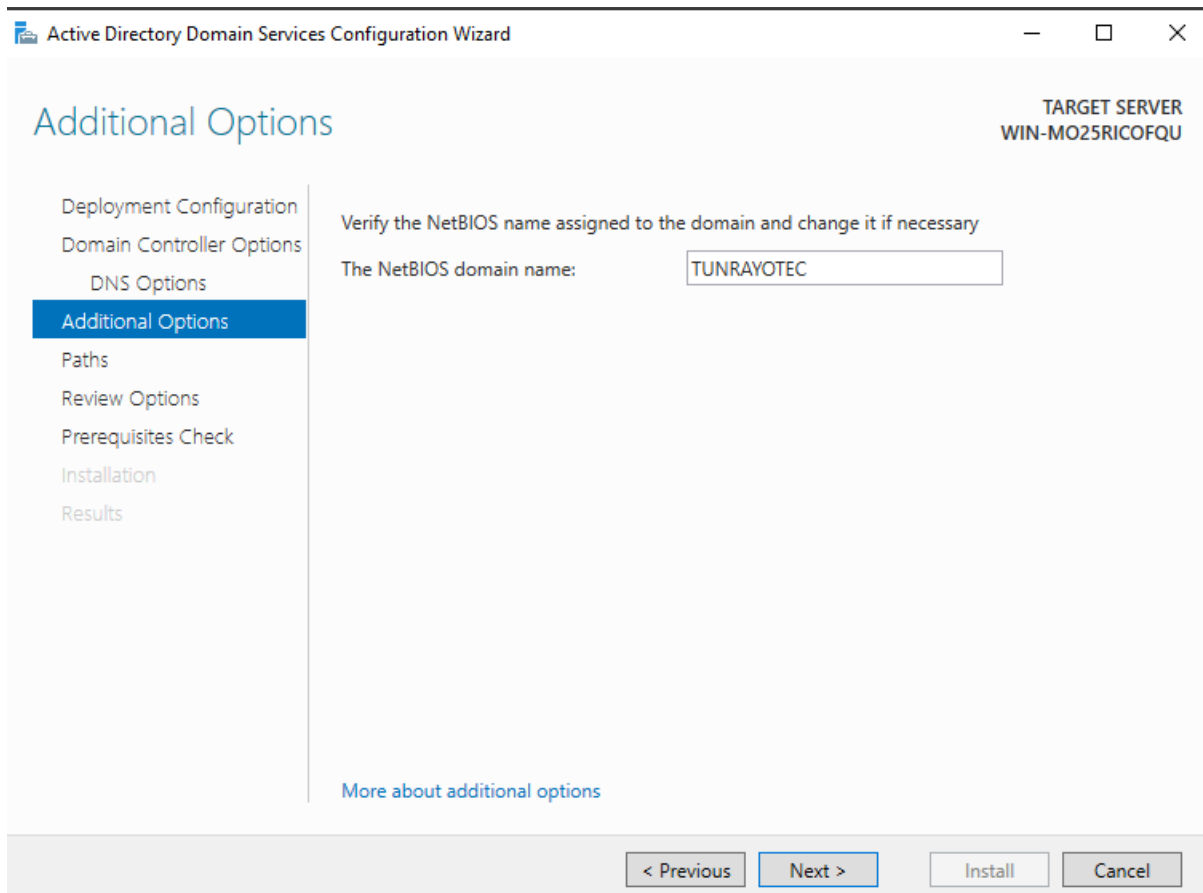


Fig 20: Click Next

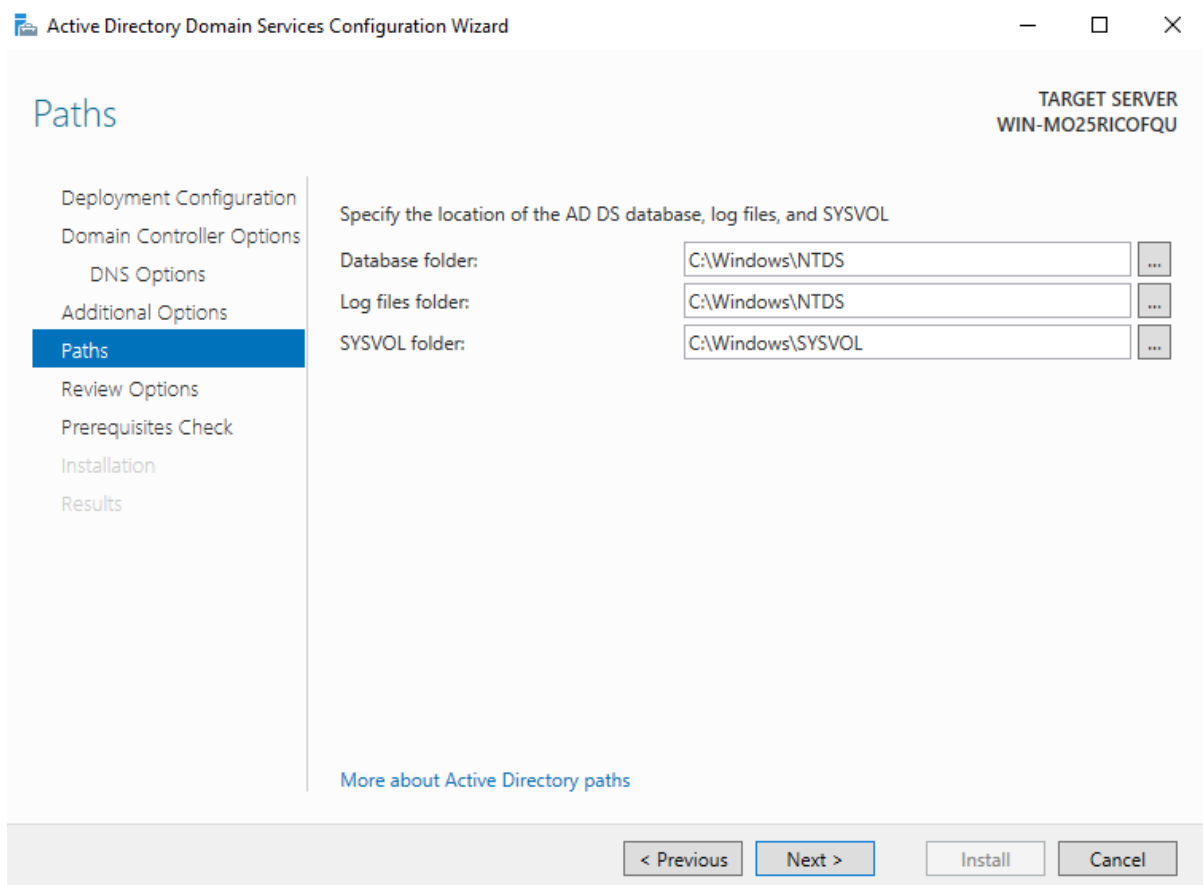


Fig 21: Click Next

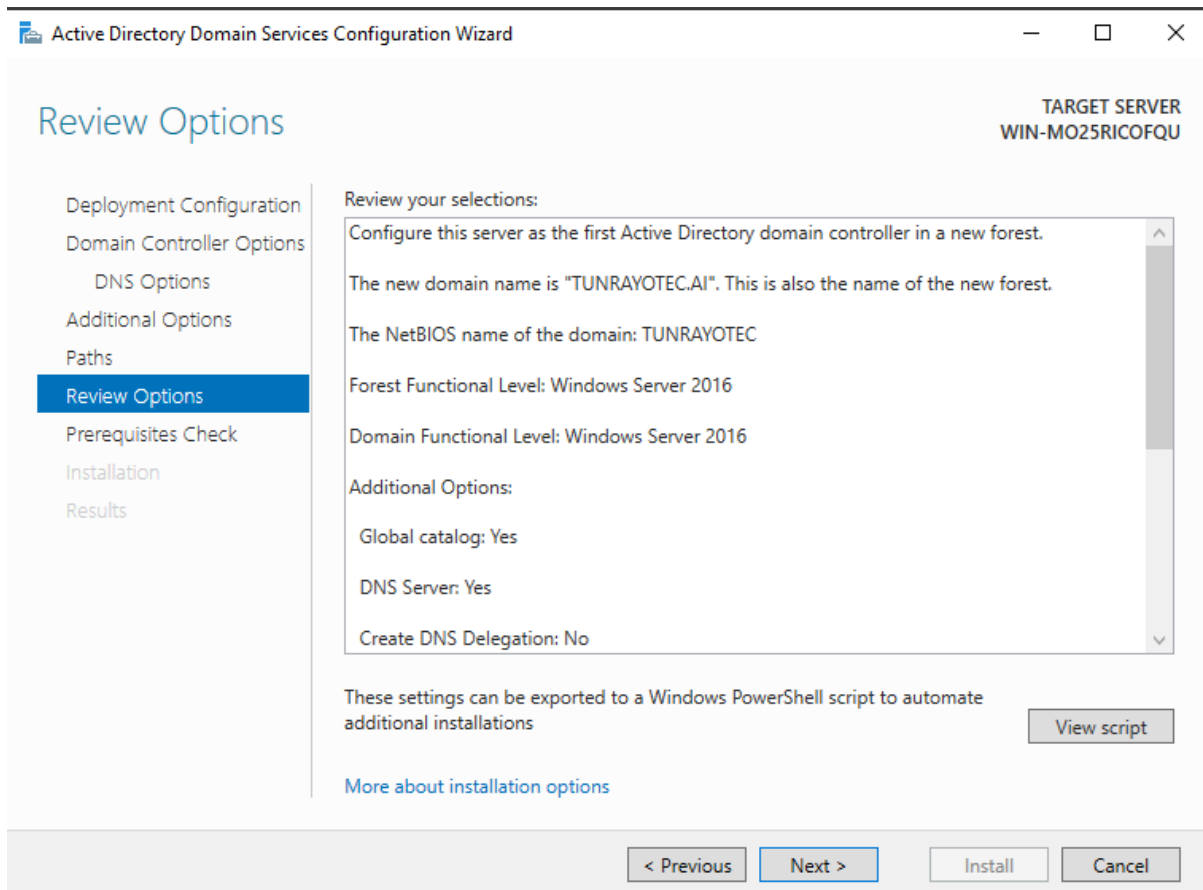


Fig 22: Click Next and click Install.

Note: After installation, the server will reboot automatically.

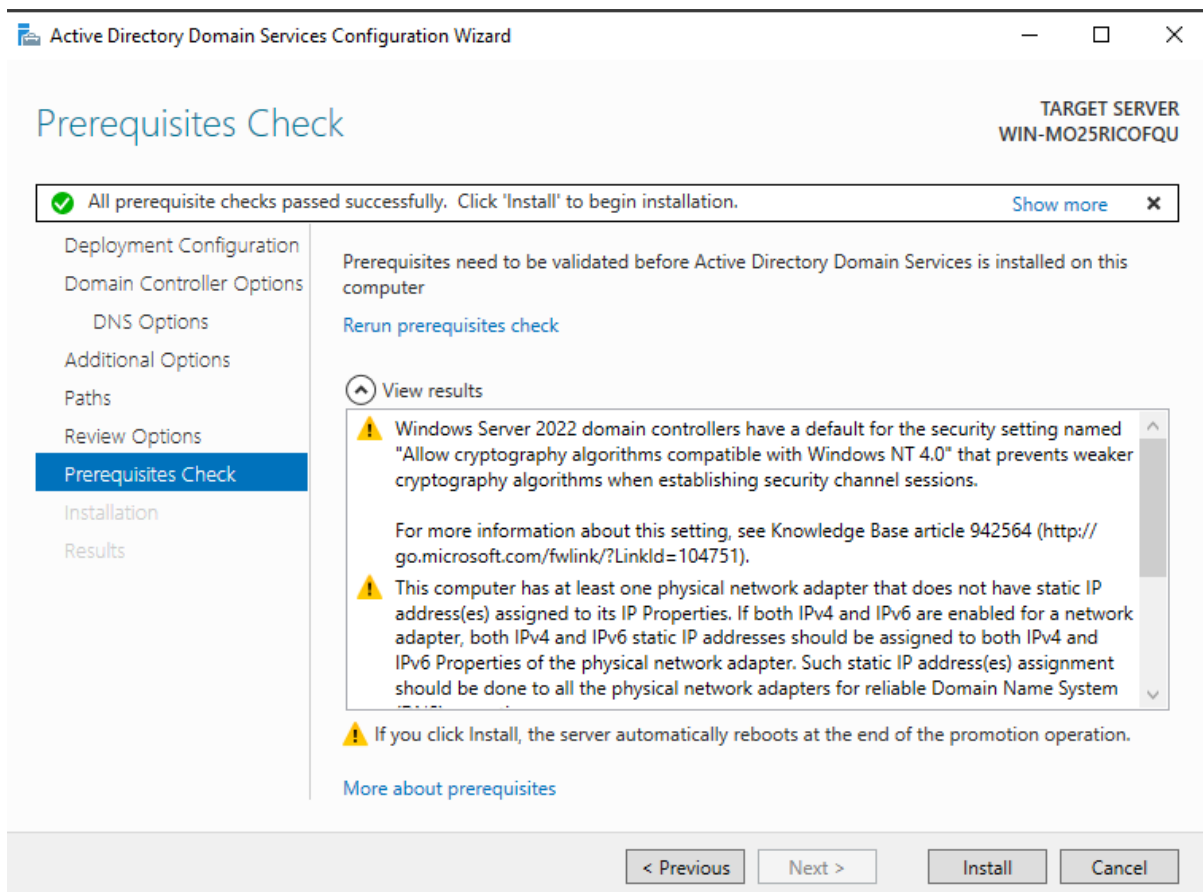


Fig 23:

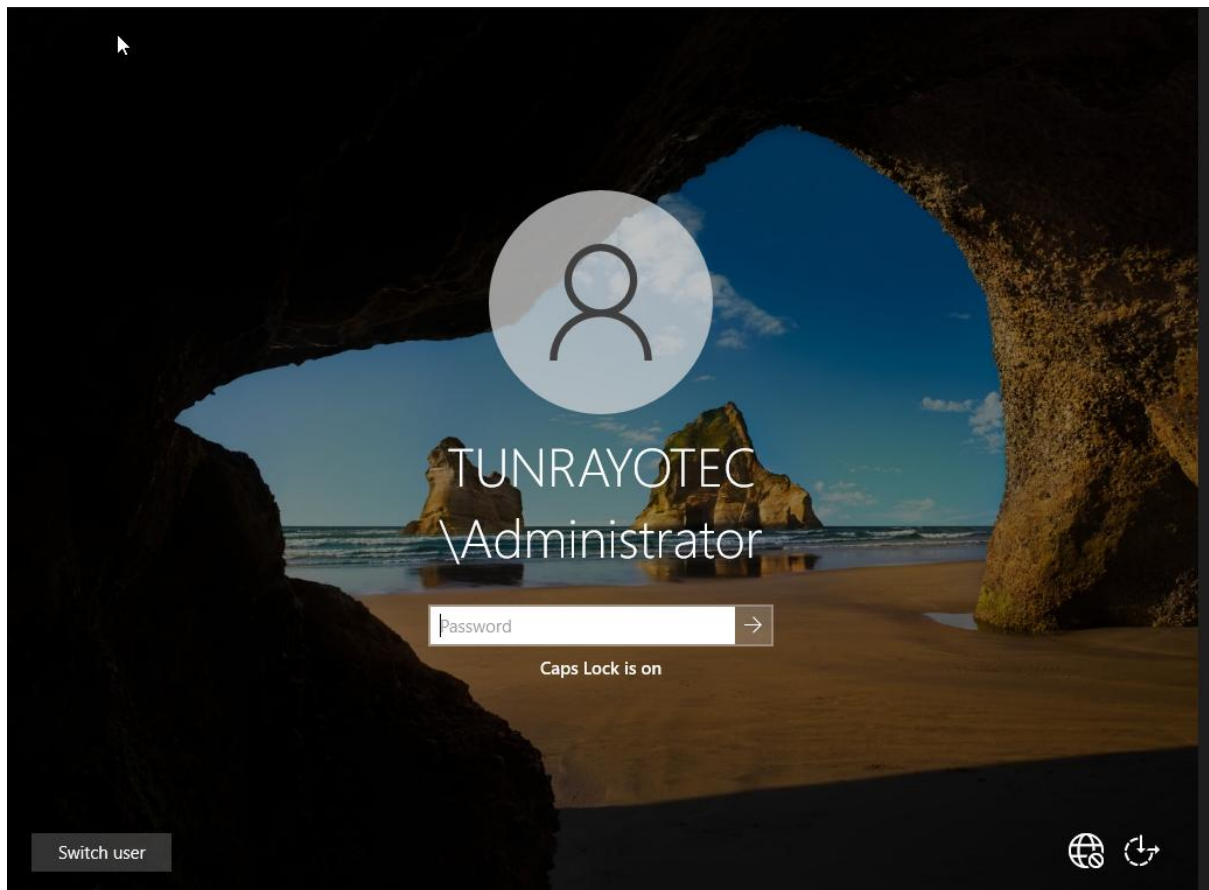


Fig 24: Navigate to Tools and select Active Directory Users and Computers

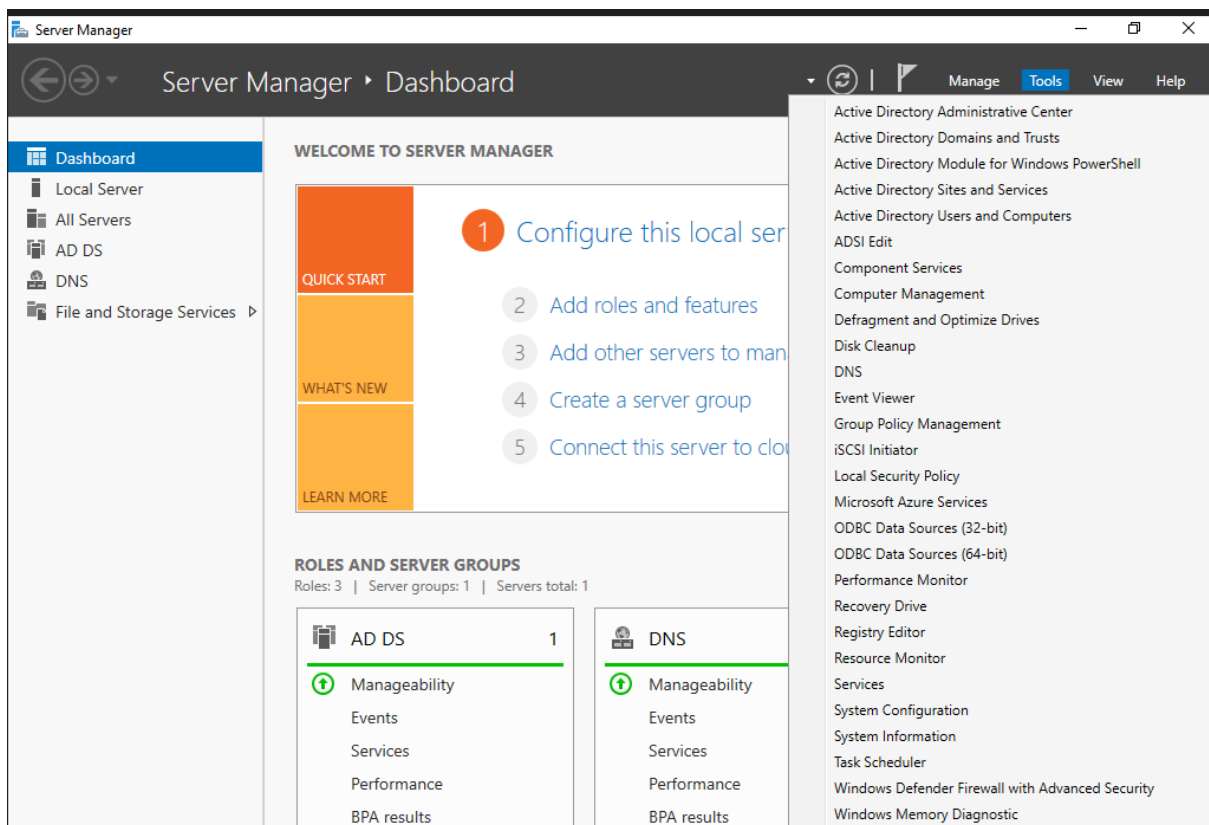


Fig 25: TUNRAYOTEC.AI Forest displayed, right click.

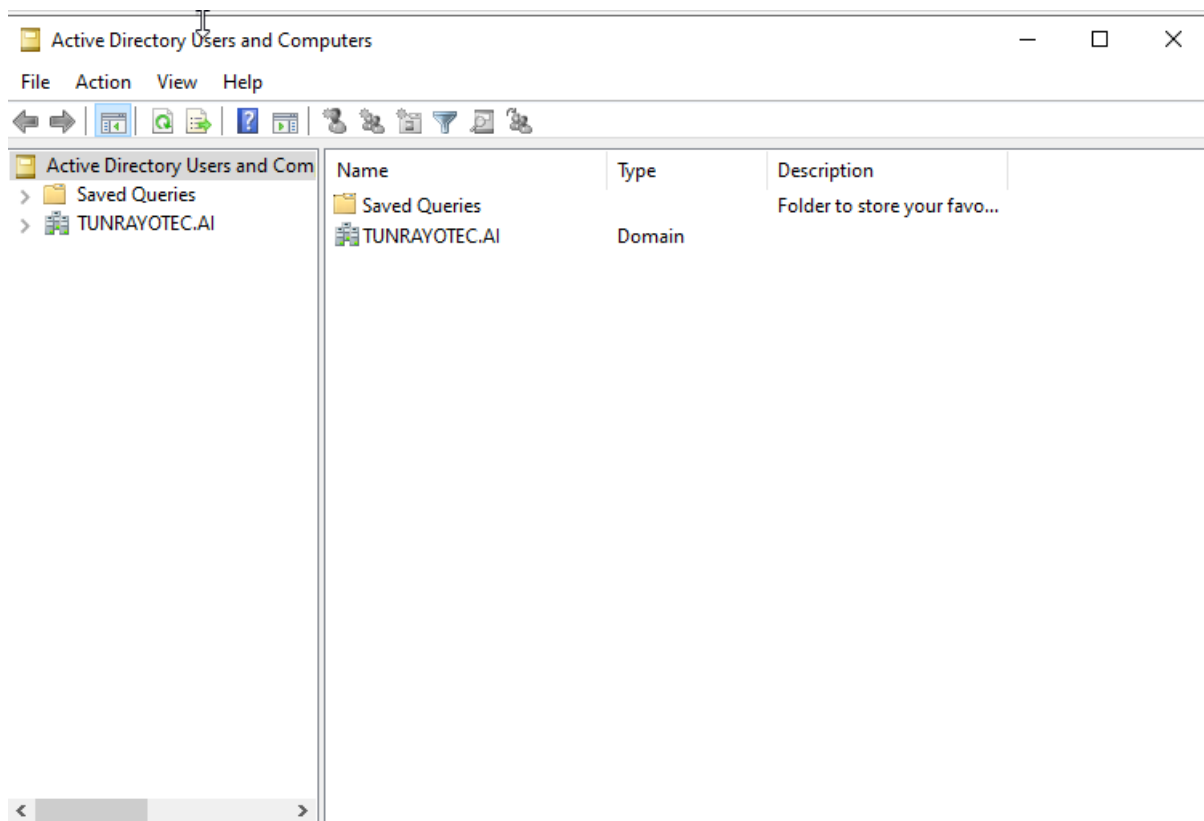


Fig 26: Click New and Organizational Unit

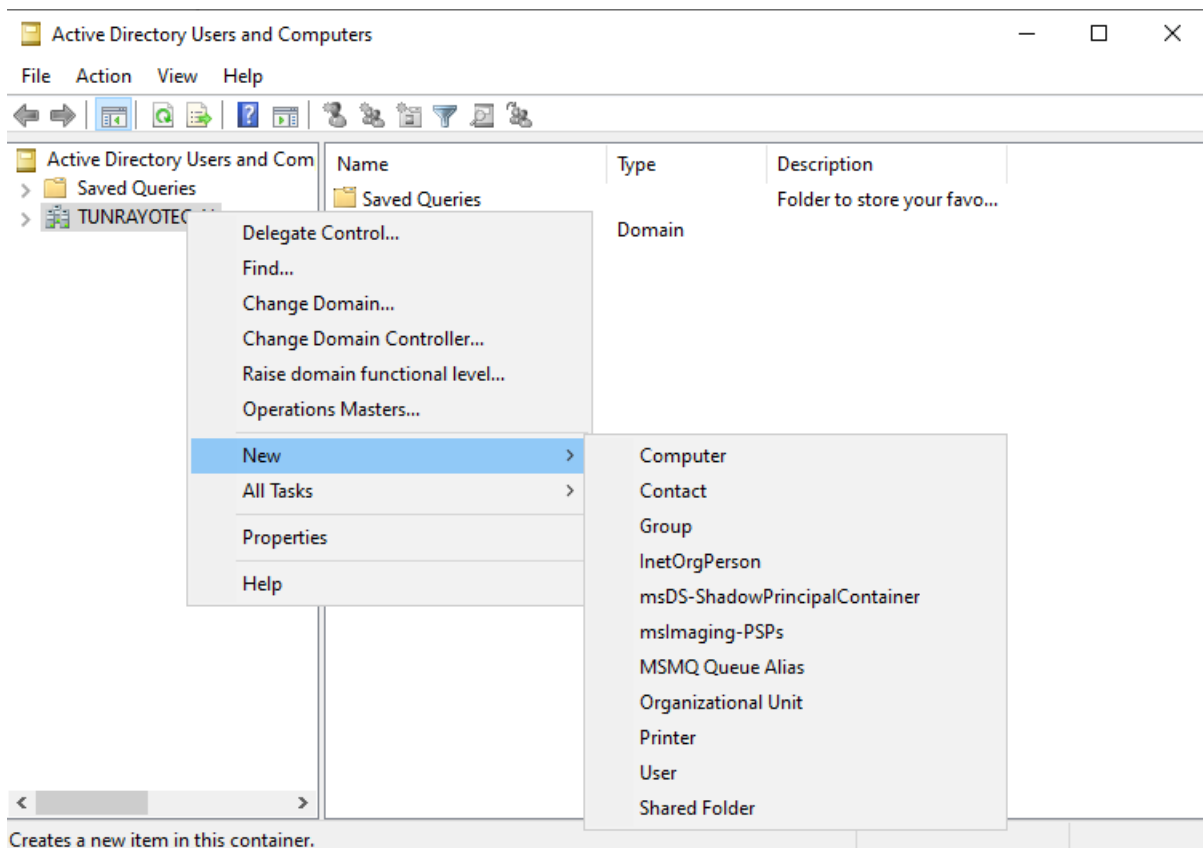


Fig 27: Create an Organisation Unit and check Protect container from accidental deletion.

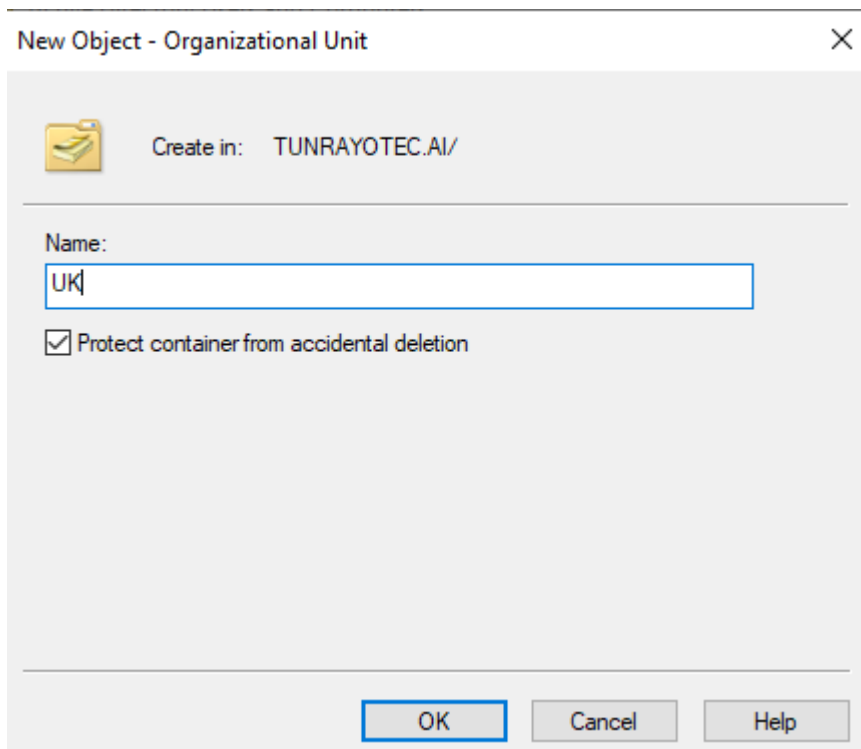


Fig 28:

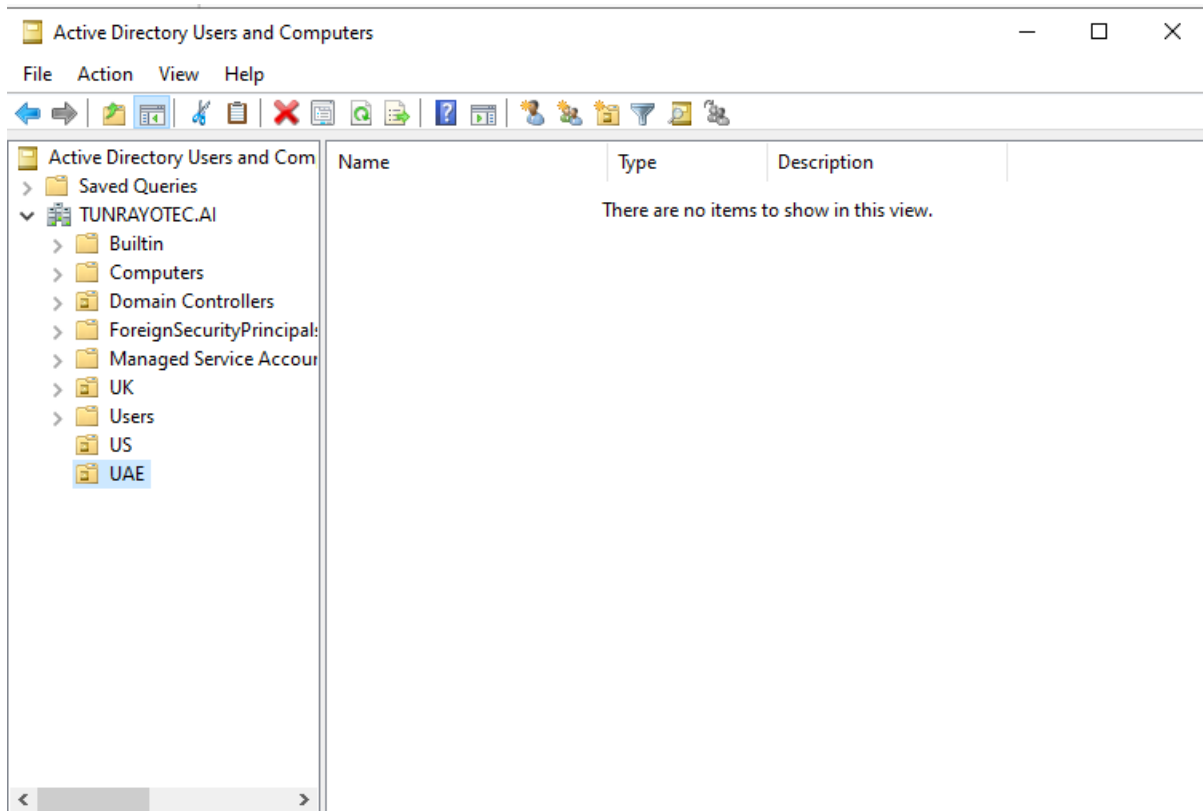


Fig 29: Create Groups, navigate to the Organisation Unit, and right click>New> Group.

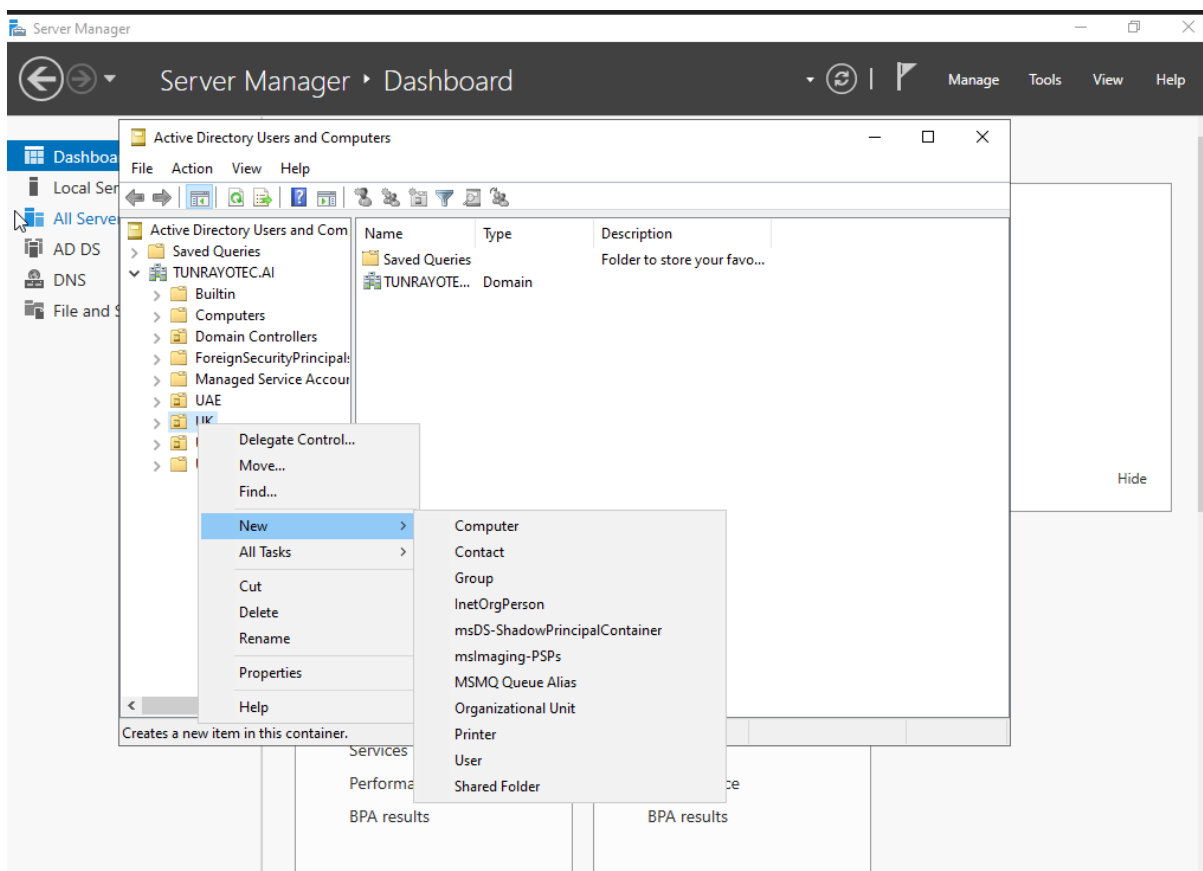
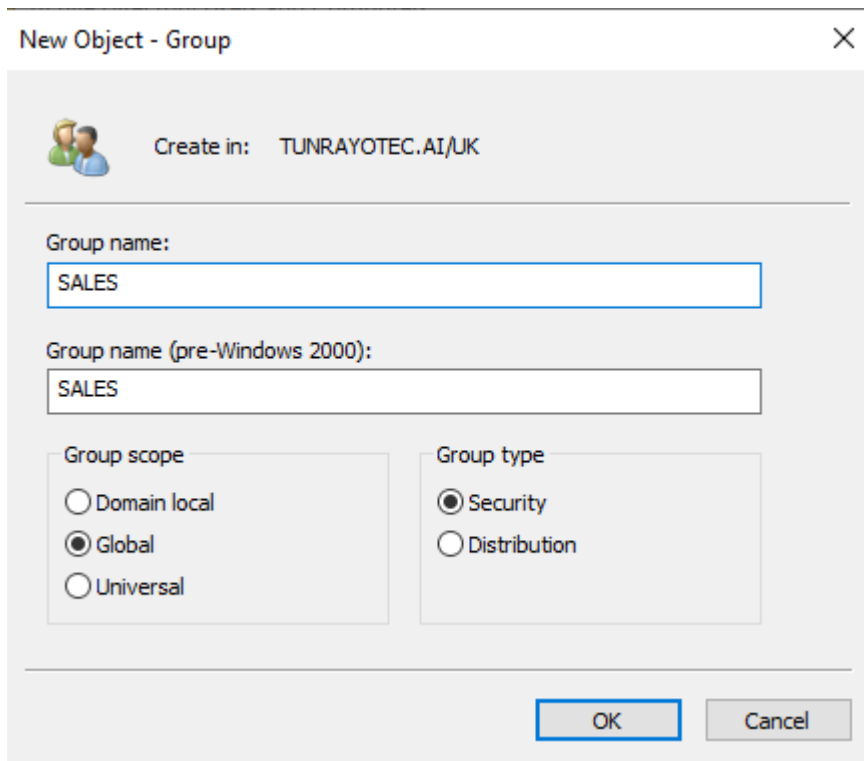


Fig 30: Enter the Group name and click OK.



New Object - Group

Create in: TUNRAYOTEC.AI/UK

Group name:
SALES

Group name (pre-Windows 2000):
SALES

Group scope

- ☐ Domain local
- ☒ Global
- ☐ Universal

Group type

- ☒ Security
- ☐ Distribution

OK Cancel

Fig 31: Create the various groups needed in each Organisation Unit.

Note: Click on the Organisation Unit and on the right the various groups should be displayed as shown in the screen shot.

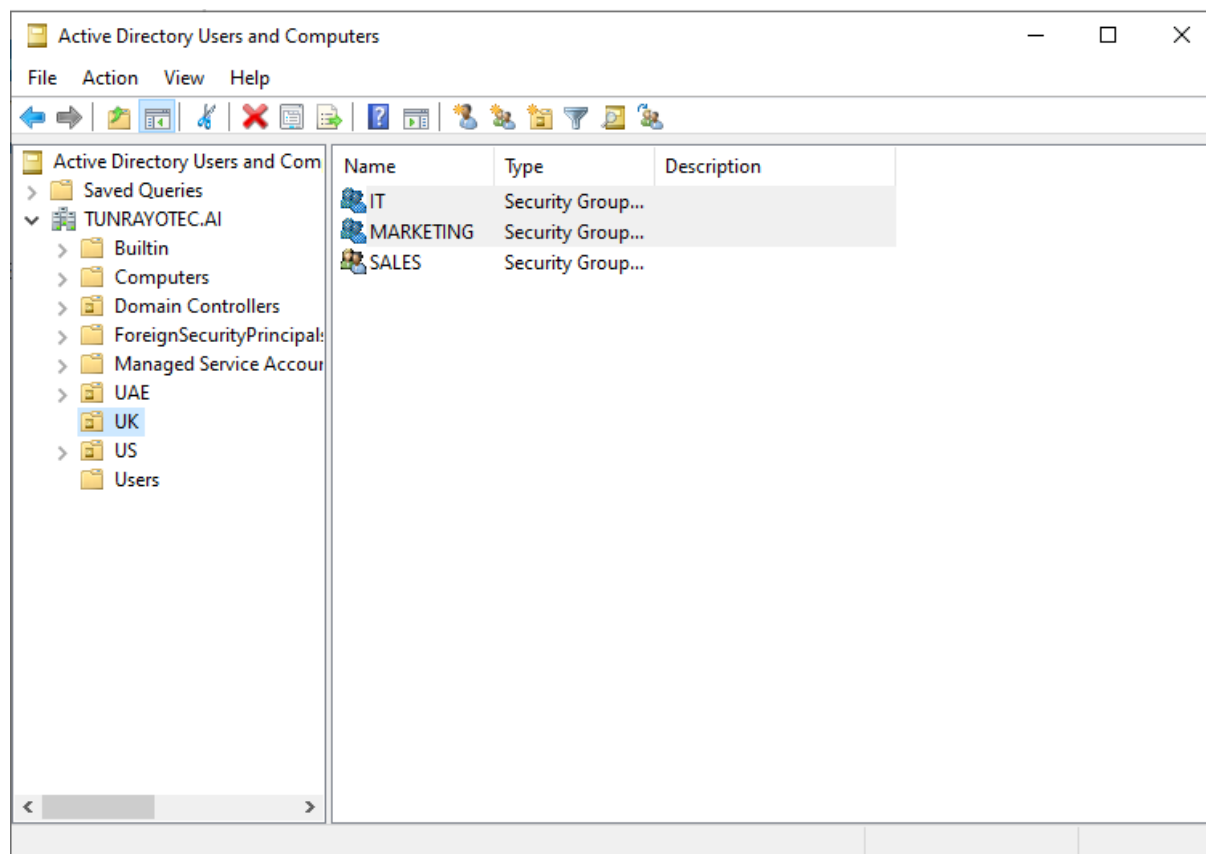


Fig 32: To create a user, click on the Organisation Unit, right click, select New and select User.

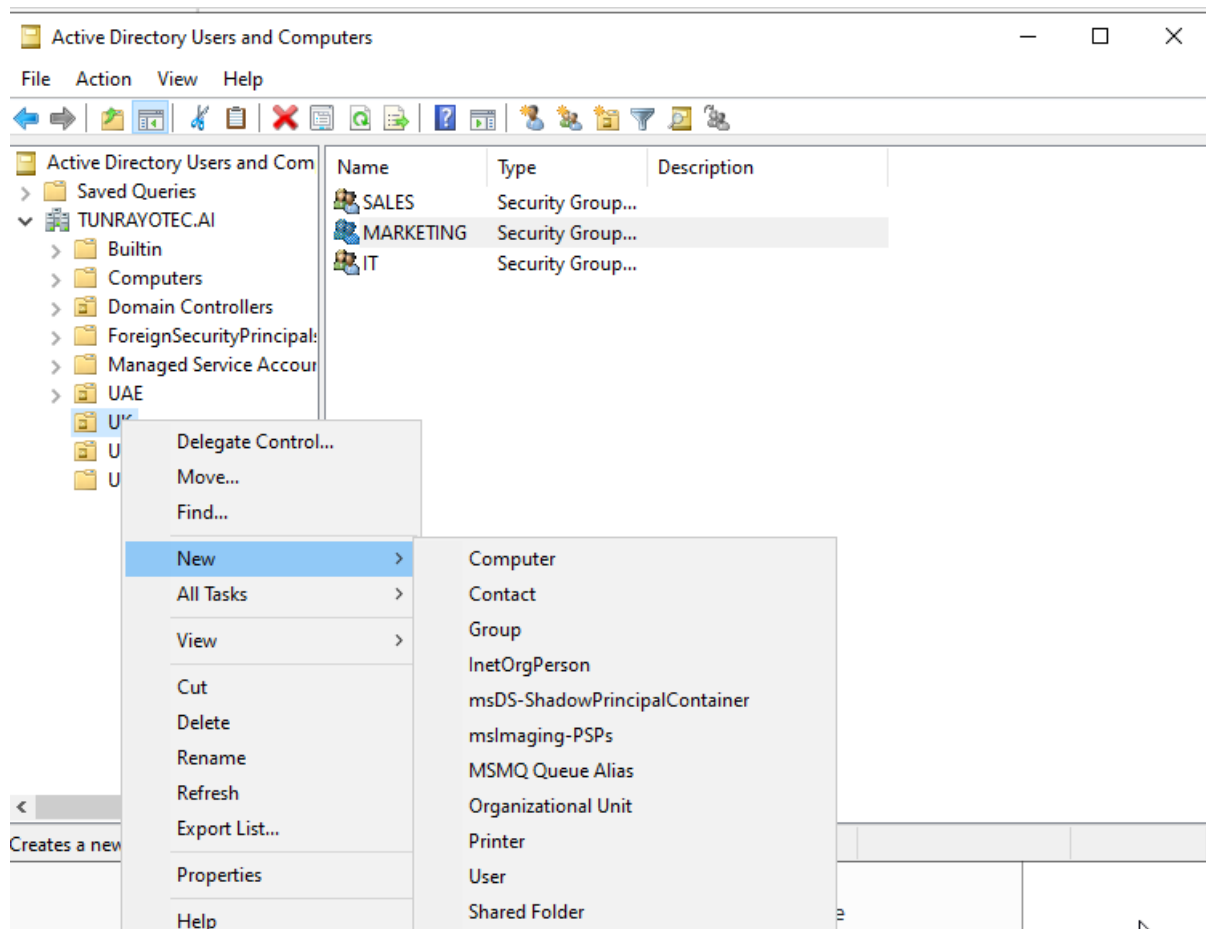


Fig 33: Enter the First name, Last name and enter the User logon name based on your company's policy.

Note: To prevent duplication ensure you use unique identifiers to differentiate users.

New Object - User ✕

 Create in: TUNRAYOTEC.AI/UK

First name: Initials:

Last name:

Full name:

User logon name:

User logon name (pre-Windows 2000):

Fig 34: Create a user password and tick the appropriate check box(es) based on the organisation password playbook /policy.

New Object - User ✕

 Create in: TUNRAYOTEC.AI/UK

Password:

Confirm password:

☐ User must change password at next logon

☒ User cannot change password

☒ Password never expires

☐ Account is disabled

Fig 35:

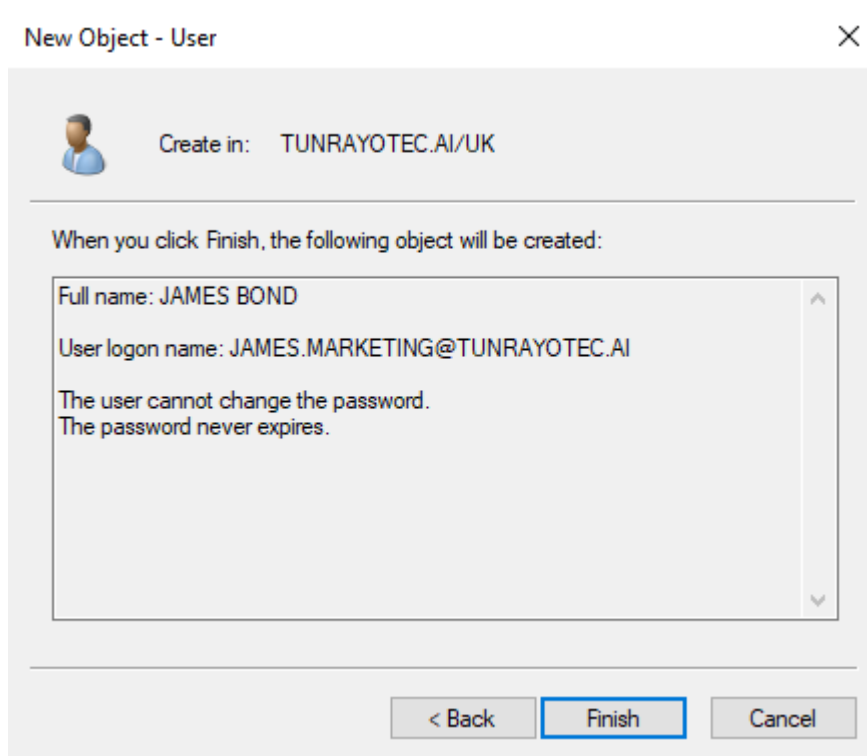


Fig 36: To add a user to a group. Right click on the specified user>New>Properties>Member of>Add>Select Group is displayed>click on Advance>Find Now >Search for the group in the search result>click OK.

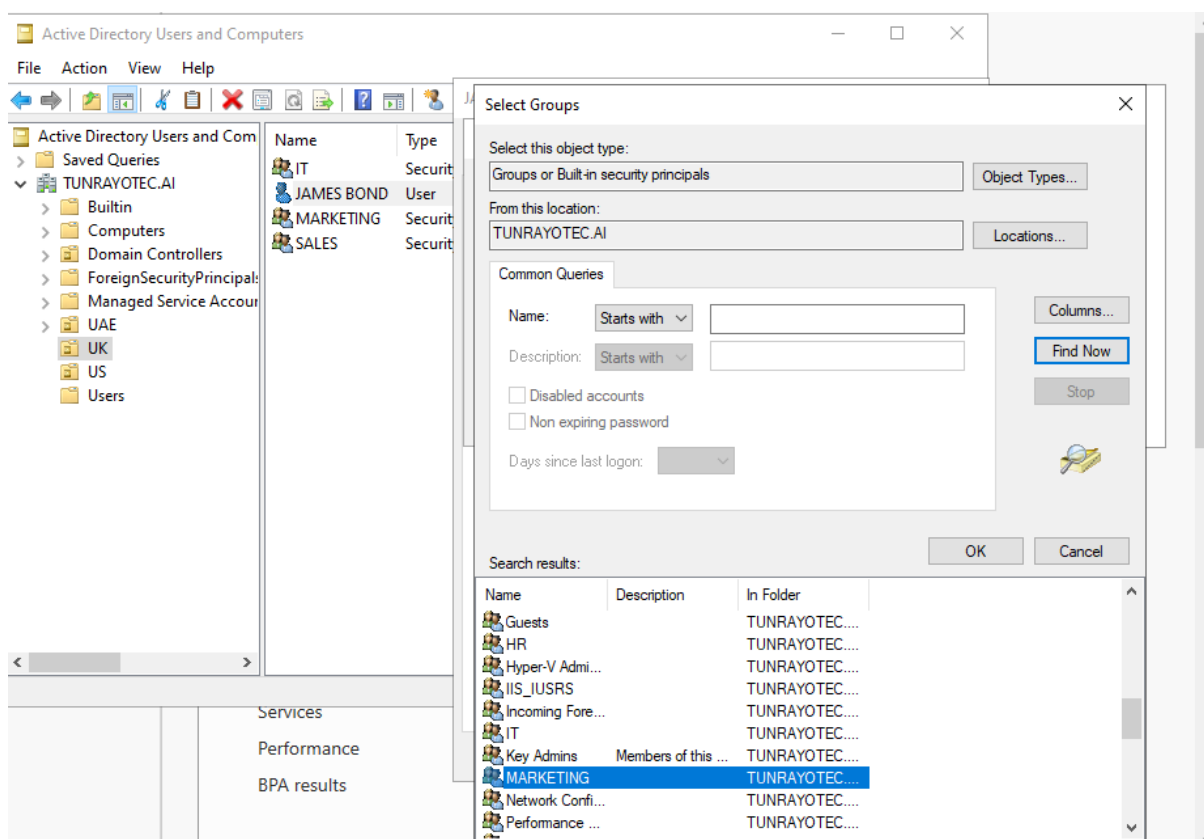


Fig 37: Click OK

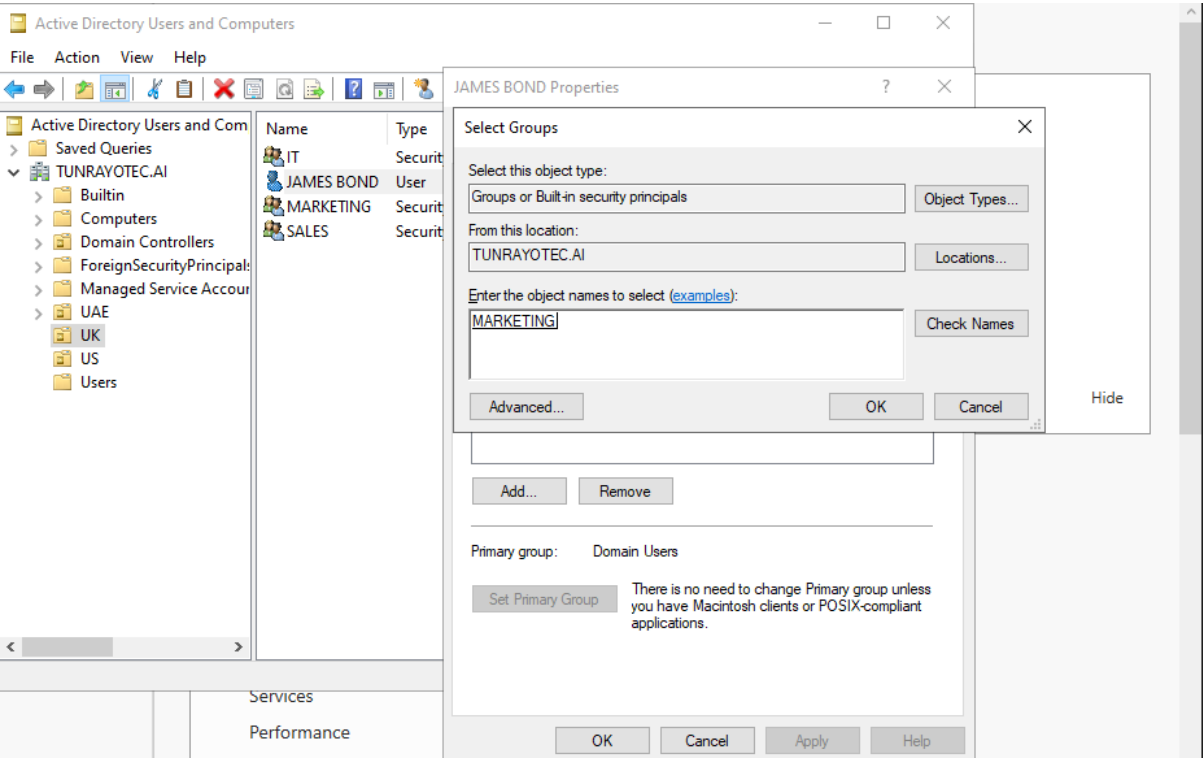


Fig 38: Click Apply

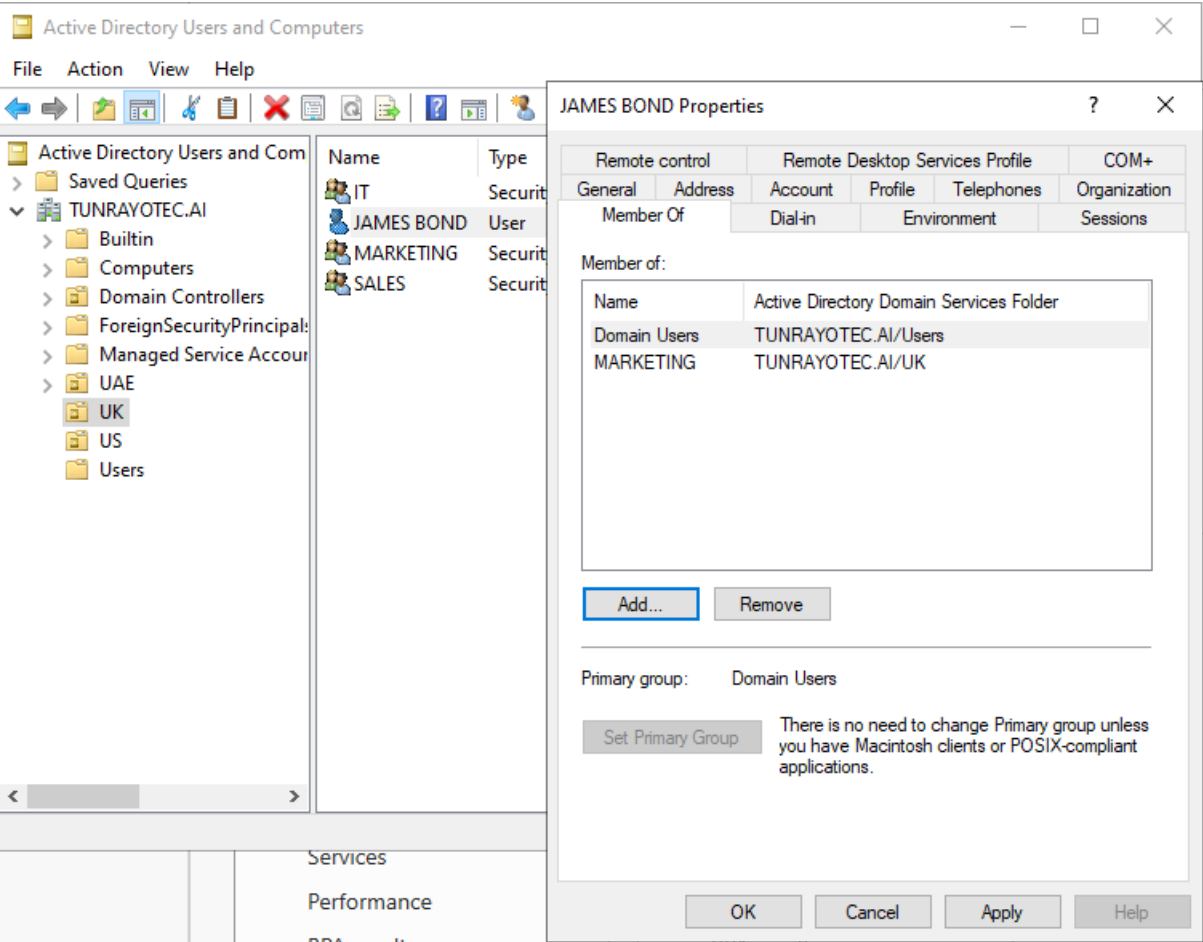


Fig 39: To verify and validate the user has been added to the specified group. Navigate to the Group>right click on the group> Properties >Members. James Bond should appear as a member of the specified group.

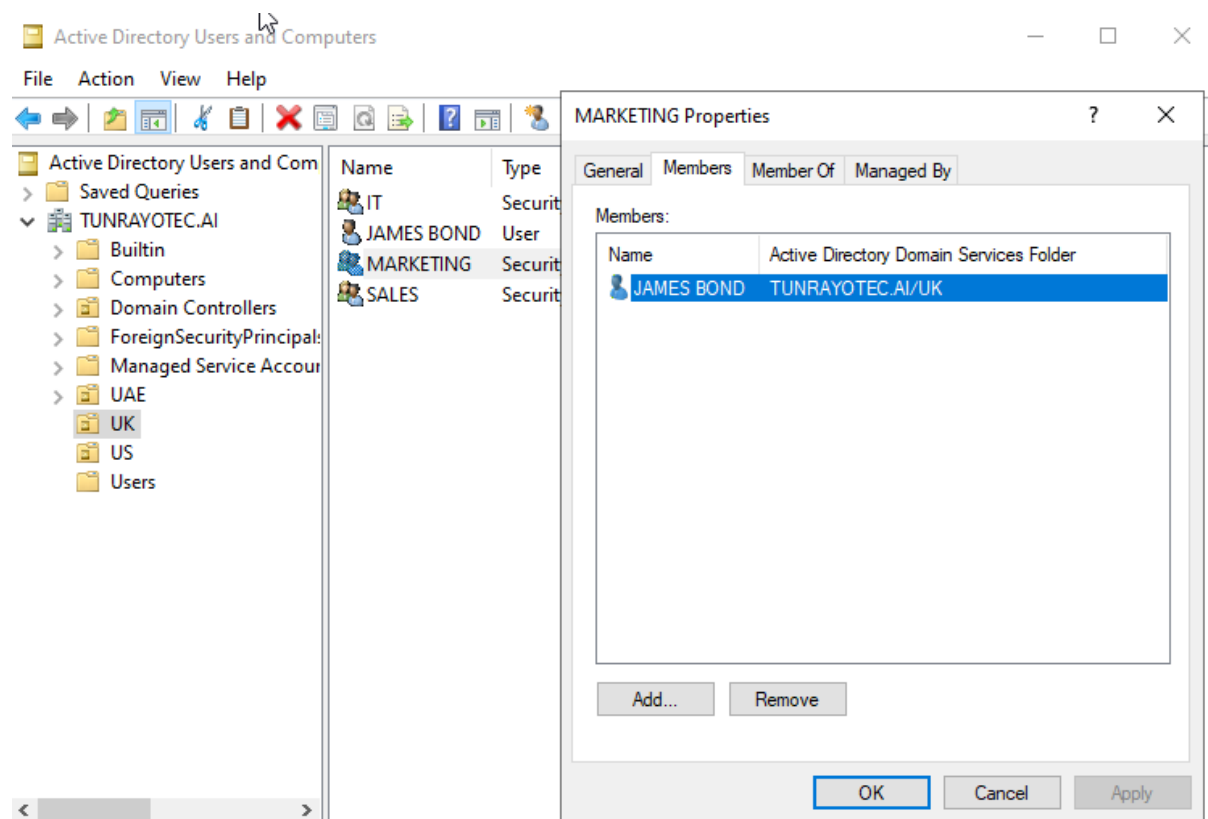


Fig 40: IP Addresses for both Server

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address. . . . . : fd17:625c:f037:2:5978:8959:e1c3:dbbb
    Link-local IPv6 Address . . . . . : fe80::5978:8959:e1c3:dbbb%4
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%4
                               10.0.2.2

C:\Users\Administrator>
```

Windows 8

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Tunrayo>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : lan
    IPv6 Address . . . . .           : fd17:625c:f037:2:f9b4:3ecd:9677:989b
    Temporary IPv6 Address . . . . . : fd17:625c:f037:2:e5d1:10bf:be9d:35bf
    Link-local IPv6 Address . . . . . : fe80::f9b4:3ecd:9677:989b%3
    IPv4 Address. . . . .             : 10.0.2.15
    Subnet Mask . . . . .             : 255.255.255.0
    Default Gateway . . . . .         : fe80::2%3
                                         10.0.2.2

Tunnel adapter isatap.lan:

    Media State . . . . .           : Media disconnected
    Connection-specific DNS Suffix  . : lan

C:\Users\Tunrayo>
```

Fig 41: In the event of having the same IP address navigate to control panel>network and internet>network and sharing Center>change advanced sharing settings>guest or public>turn on network discovery>turn on file and printer sharing>save.

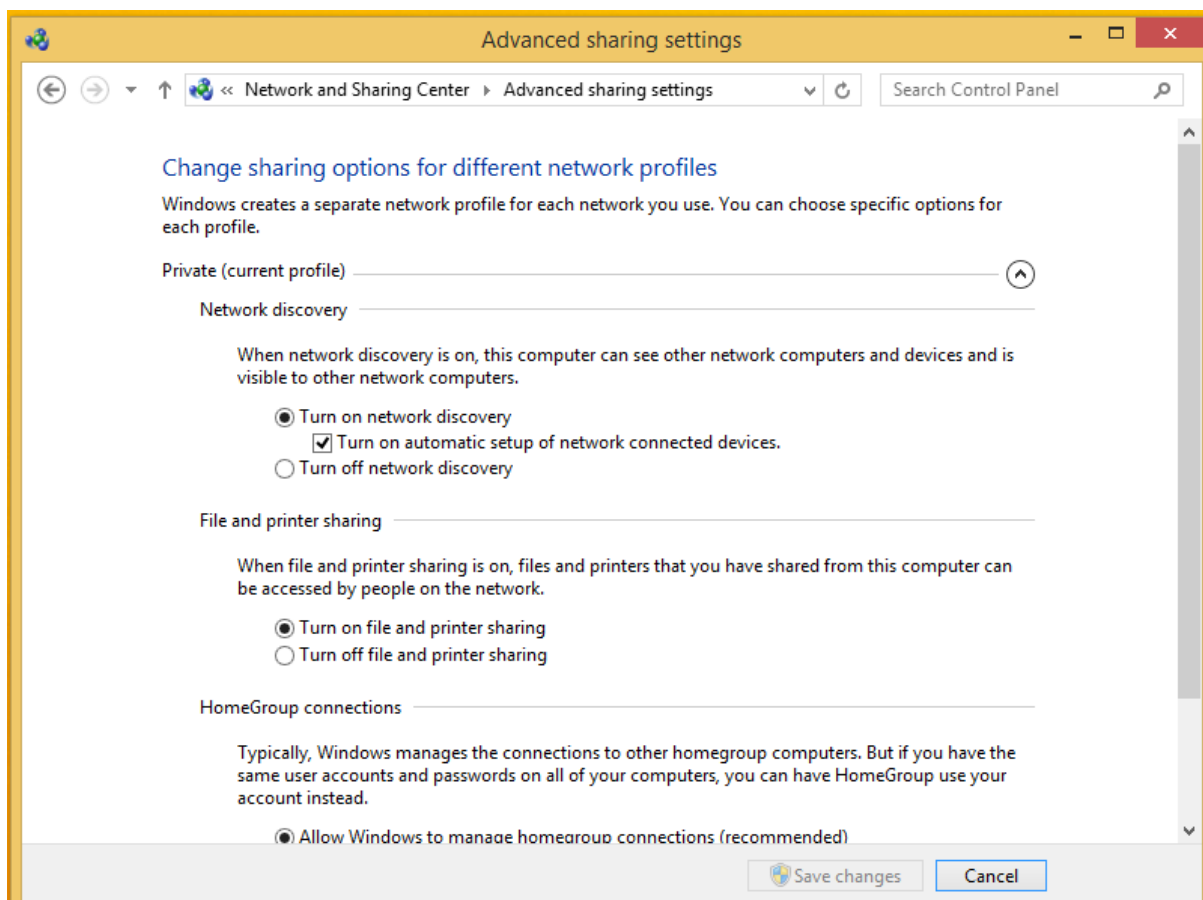


Fig 42:

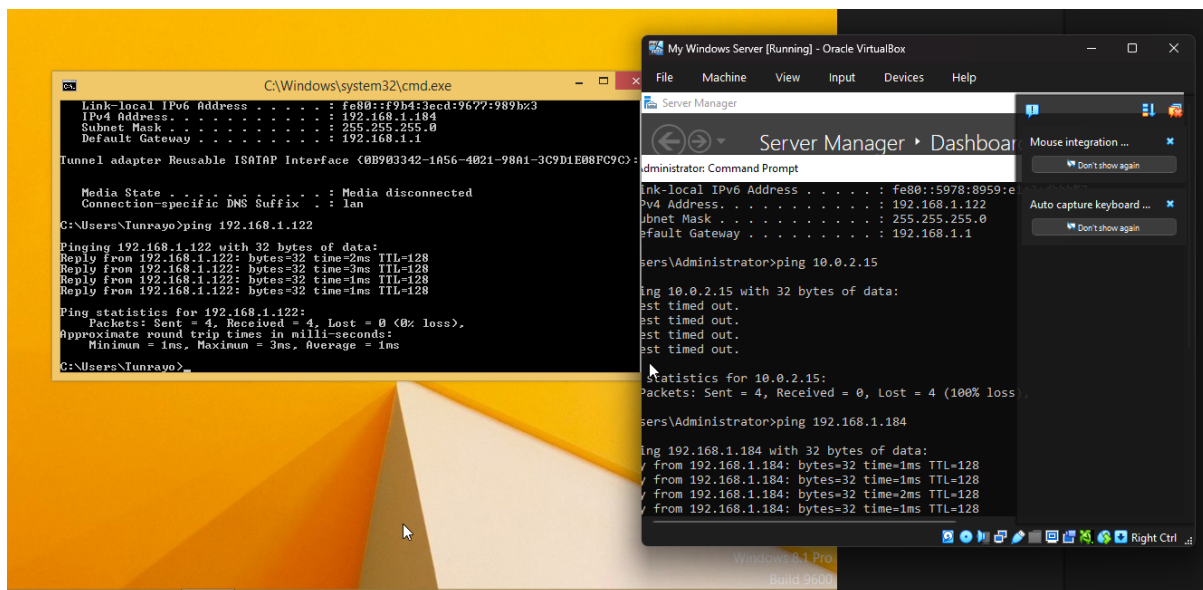


Fig 43(a): Navigate to the PC that will be issued to the staff i.e. Windows 8, use Windows+R, enter control>navigate to Network and Internet>Network and Sharing Center>Ethernet> Ethernet Properties>click on Internet Protocol Version 4(TCP/IPV4)>Properties>here you need to ensure the DNS is set to the IP of the Server(this is to ensure Windows 8 is getting all its information from the server. The server will be controlling it)> and enter the details for the Preferred DNS server IP Address and click OK and close.

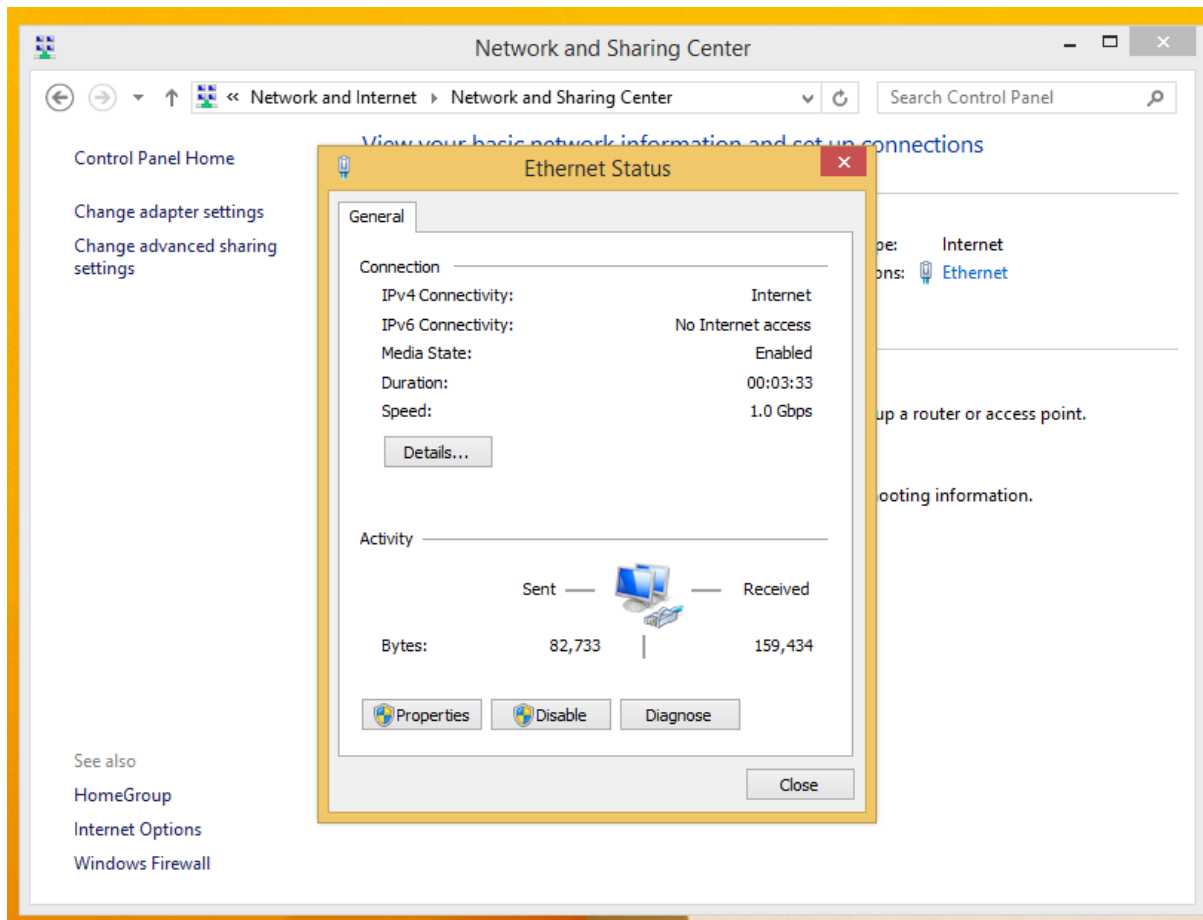


Fig 43(b)

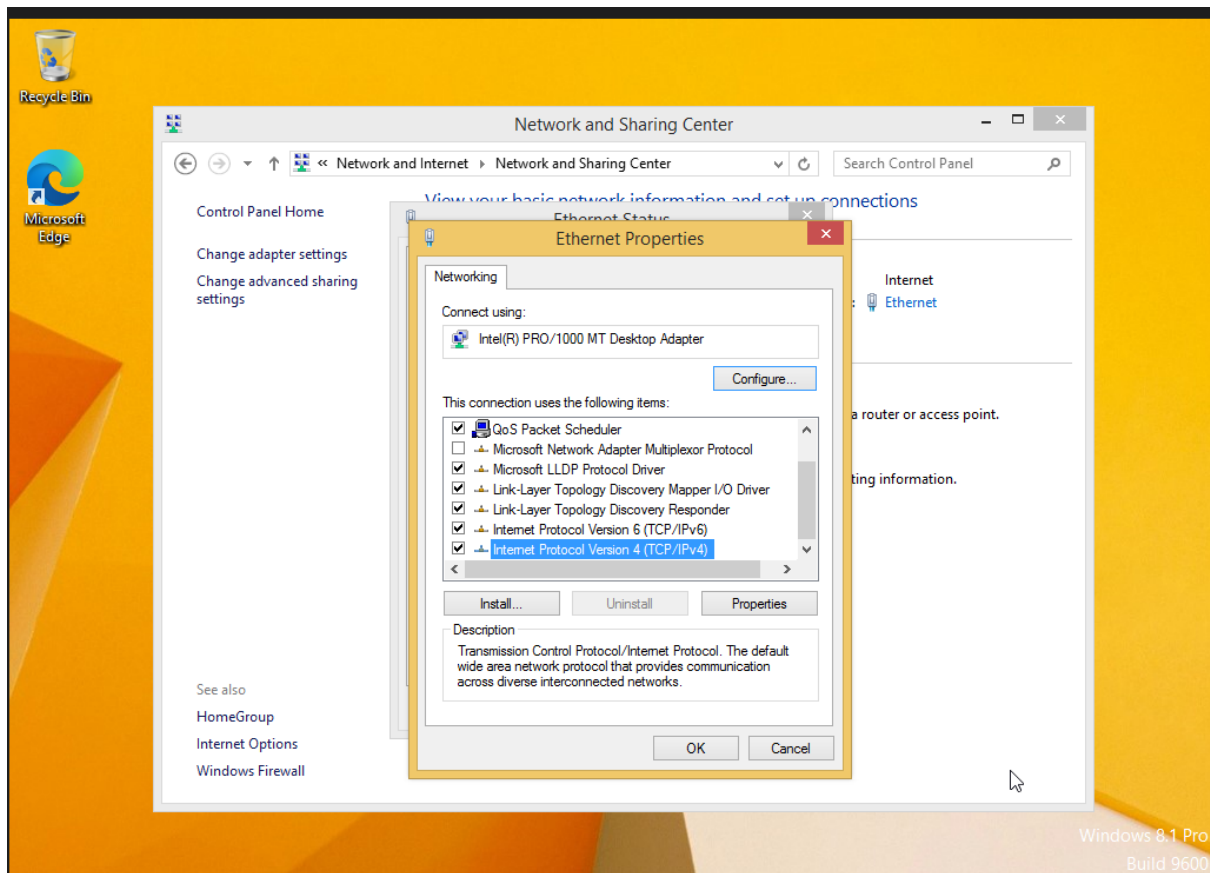


Fig 43 ©:

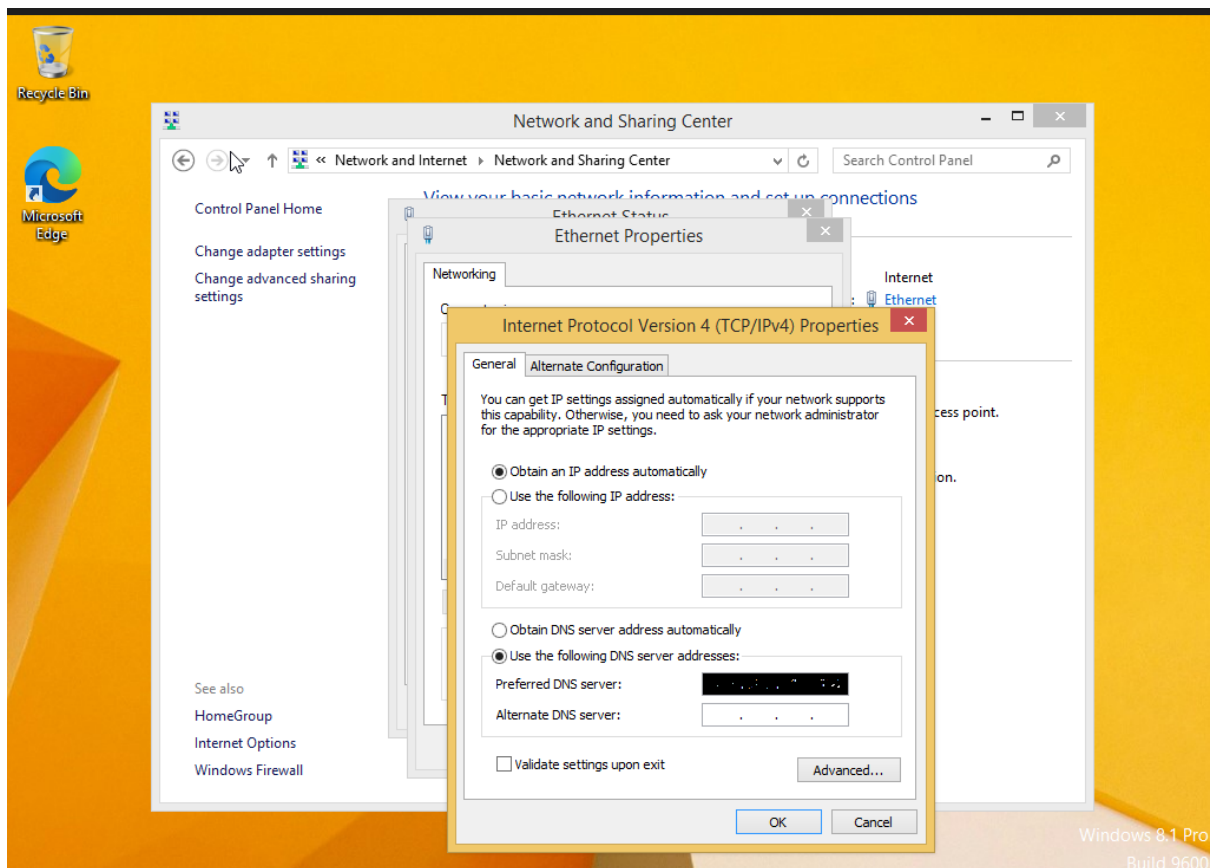


Fig 45(a): Navigate to the File Explorer, right click on System and Security>System Protection>Computer Name>click change to rename the computer or change domain or workgroup

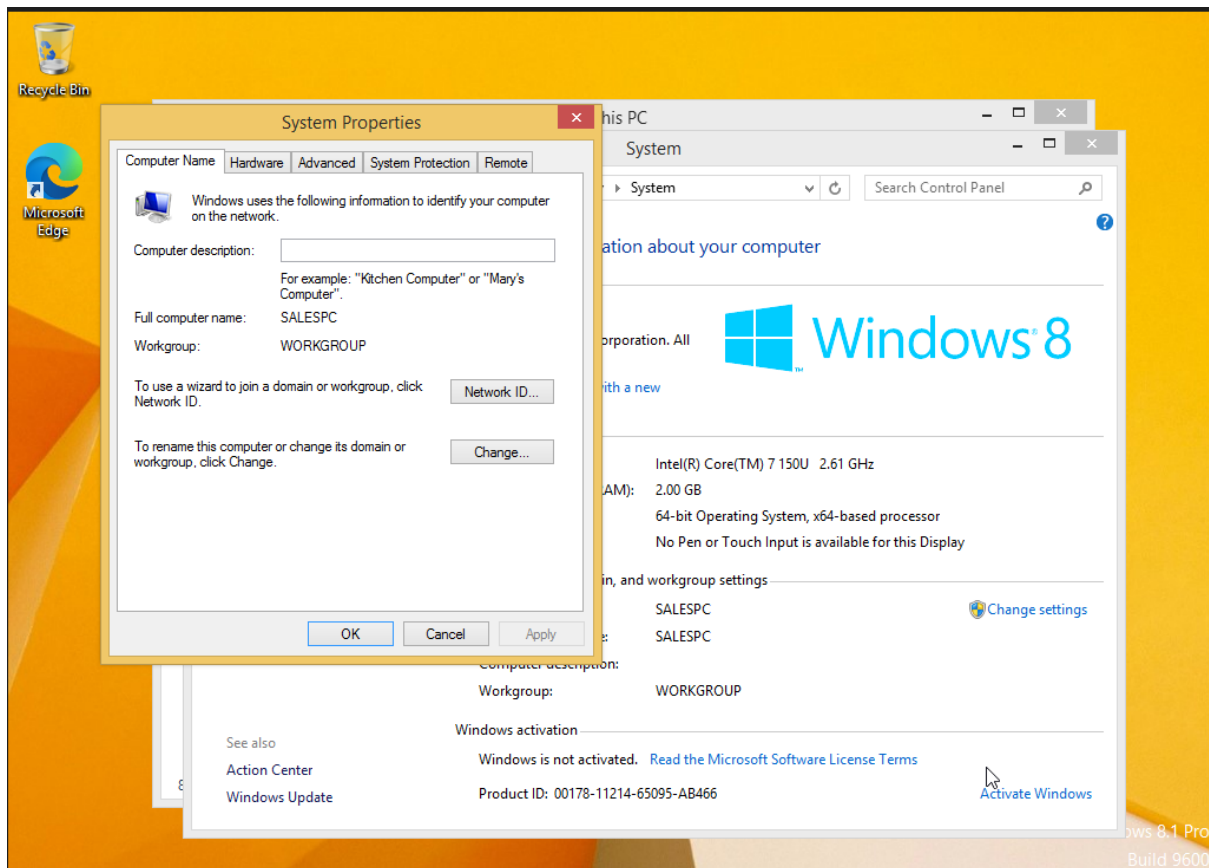


Fig 45(b): Enter the domain name, click OK.



This PC
System

Control Panel > System and Security > System

Search Control Panel

Control Panel Home

- Device Manager
- Remote settings
- System protection
- Advanced system settings

View basic information about your computer

System Properties

Computer Name/Domain Changes

You can change the name and the membership of this computer. Changes might affect access to network resources.

Computer name:
SALESPC

Full computer name:
SALESPC

More...

Member of

☒ Domain:
TUNRAYOTEC.AI

☐ Workgroup:
WORKGROUP

OK Cancel

OK Cancel Apply

Change settings

Activate Windows

Windows 8.1 Pro
Build 9600

Fig 45 ©: Enter the

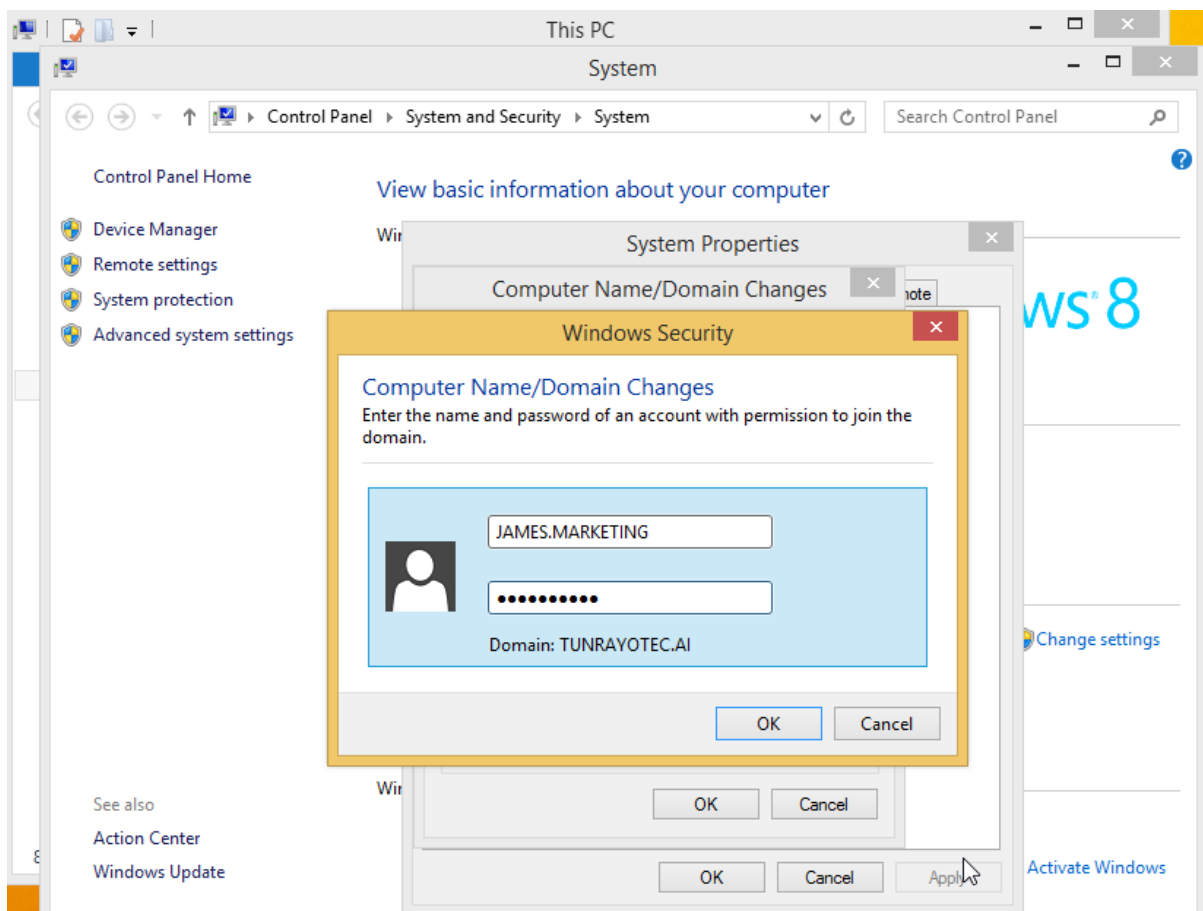


Fig 45(d)

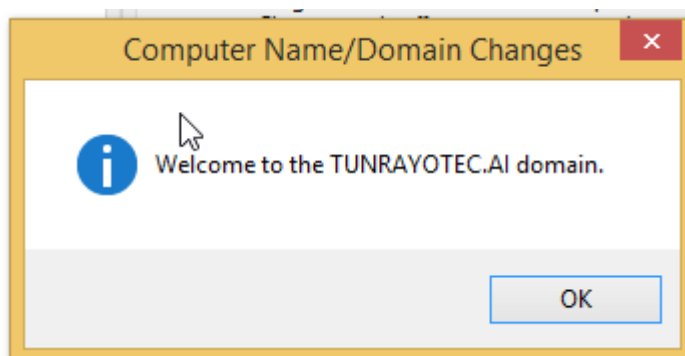


Fig 46 (a): To configure Least Privilege permissions to a user, navigate to> the server>tools> group policy management>click>expand the Forest>expand the domain>group policy objects>right click on new>enter the name of the policy.

Note: This is to assign permissions to certain group, individual, or individuals. Take for example if we have ten members in the marketing team, it makes no logical reason to create policies for each a member is added to a group. The best and efficient way is to have a benchmark policy for diverse groups. In the event we have new employees, all we need to do is attach the relevant policies to them)

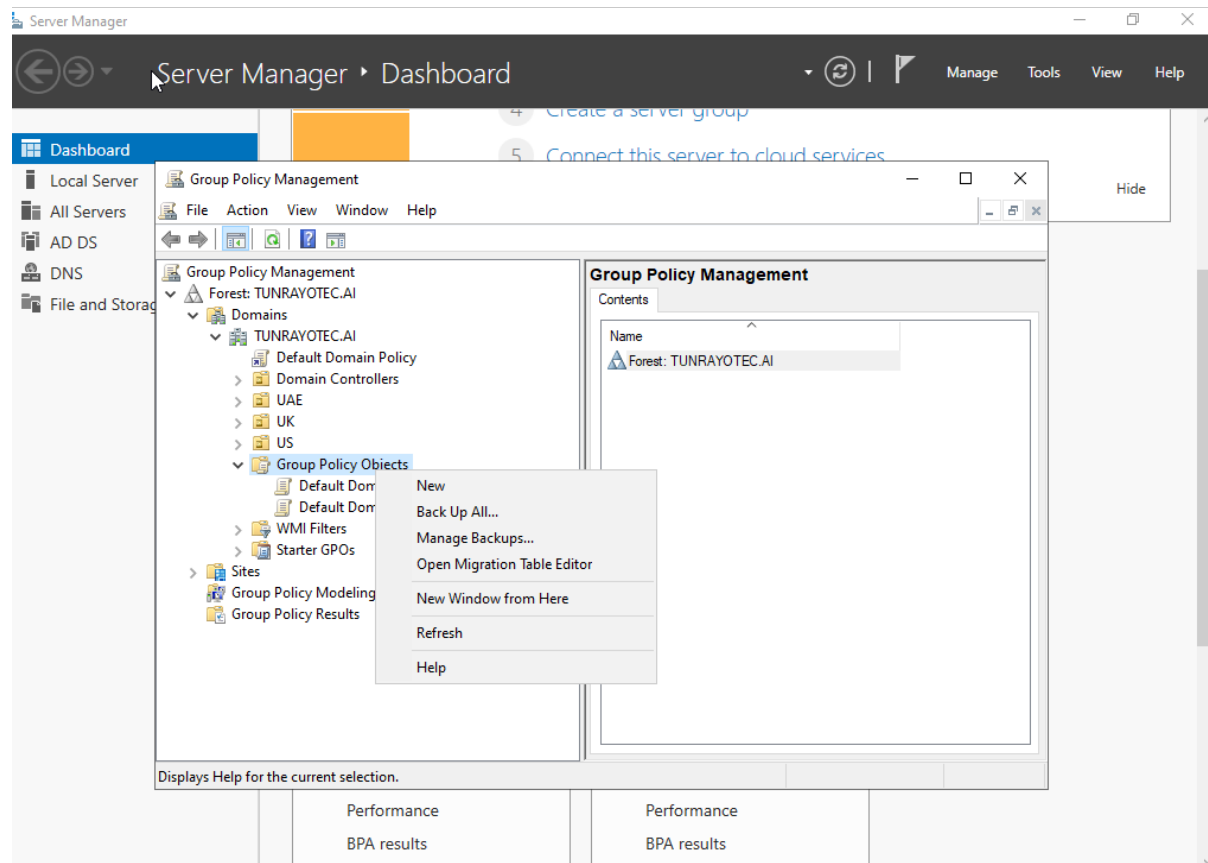


Fig 46(b)

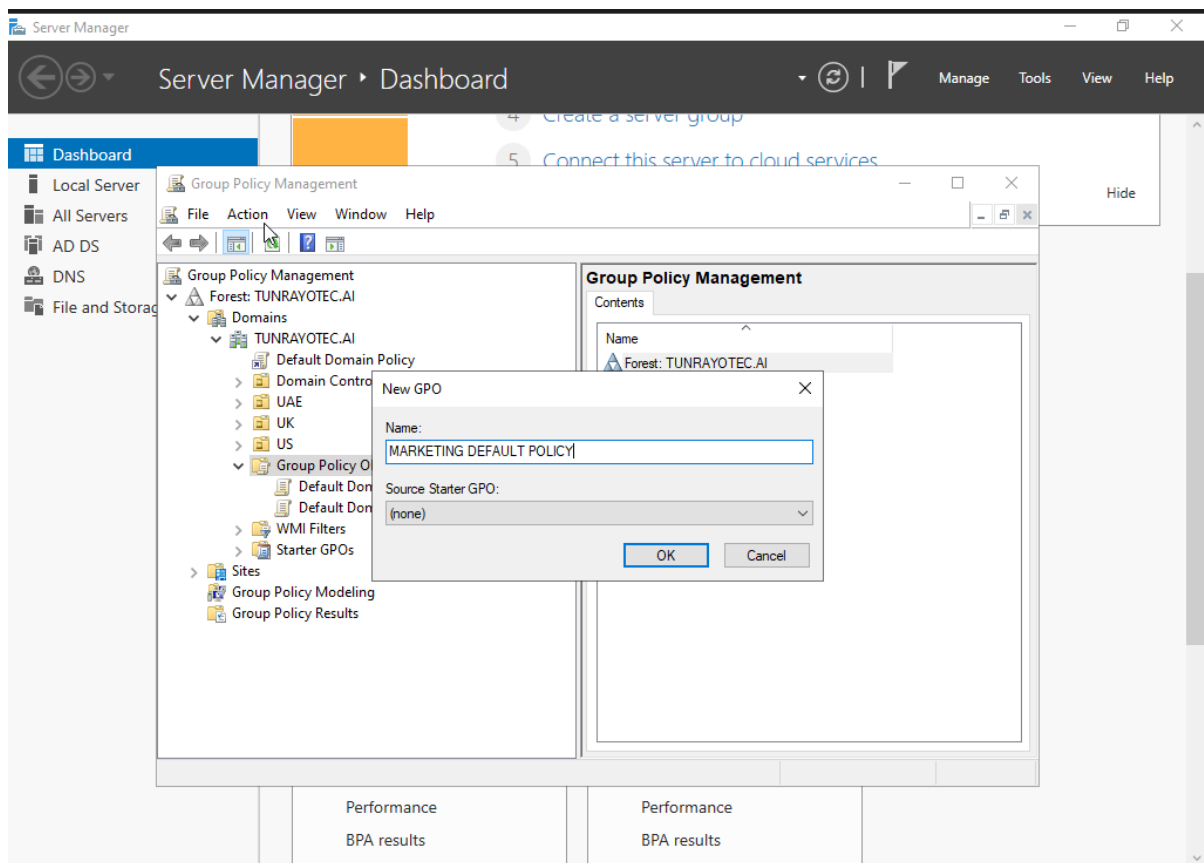


Fig 46©: The default policy is added to the Group Policy Objects

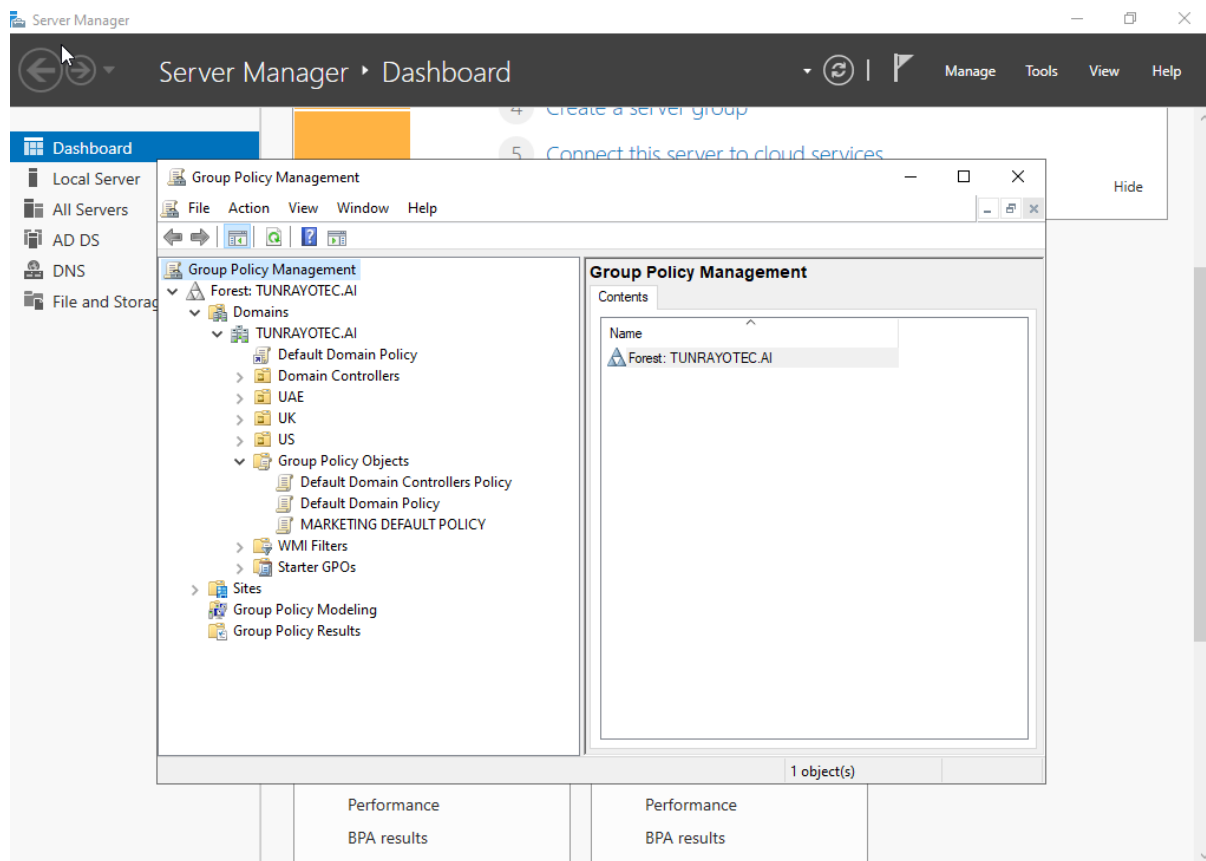


Fig 47: Right click on the Marketing Default Policy> edit>

Note: There are two types of configurations namely:

- **Computer Configuration:** This configuration is based on the control settings that affect the entire computer, no matter who logs in.
This is also applied at Machine Level and System start up.
- **User Configuration:** This configuration controls settings specific to an individual person using the system.
This is applied at pre-user account level and at User login.

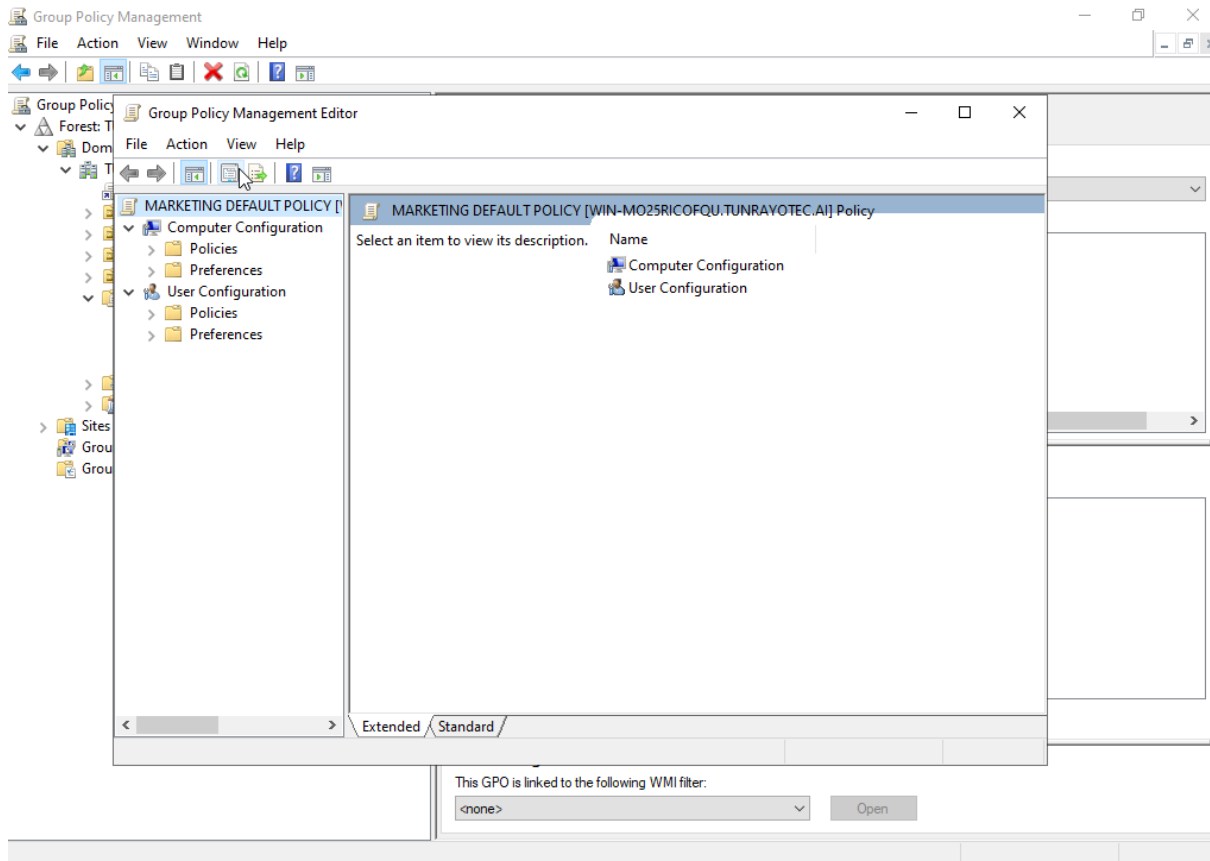


Fig 48: Select the Computer Configuration>Windows Settings>Security Settings>password policy>double click on enforce password history> enter details>click apply>OK.

Note: Password History is a security feature that remembers your previous passwords so you cannot keep reusing the same one repeatedly, until one has exhausted the defaulted number.

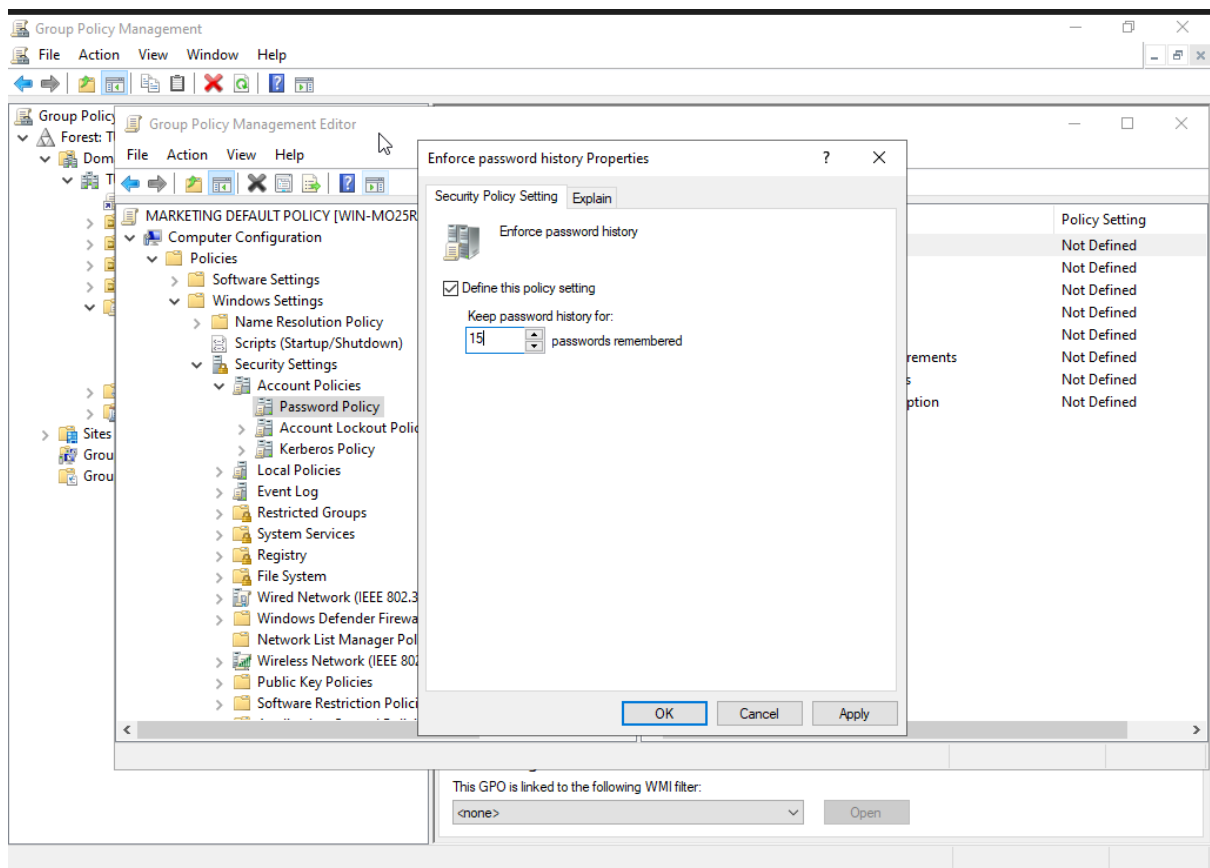


Fig 49: To prohibit the user to Control Panel, navigate to > User Configuration>

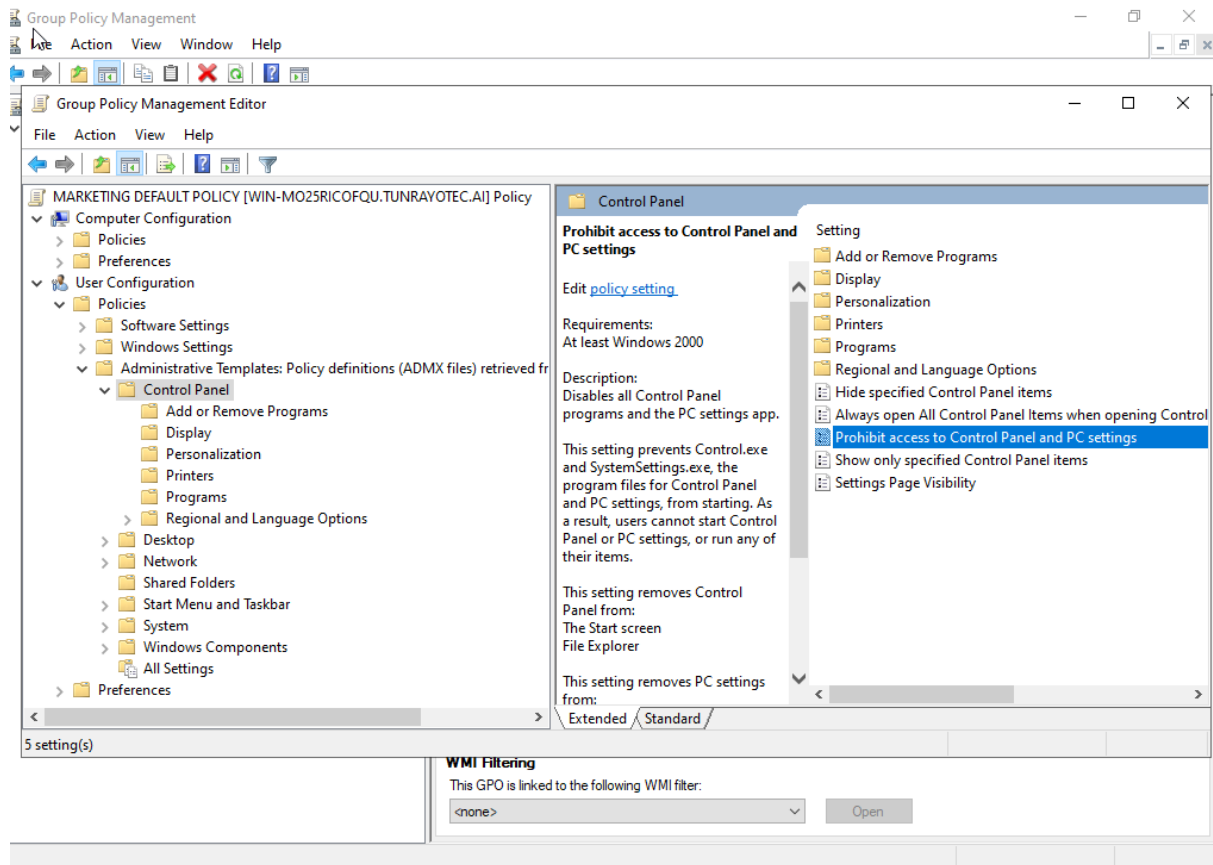


Fig 50: To prevent user from using the command prompt .

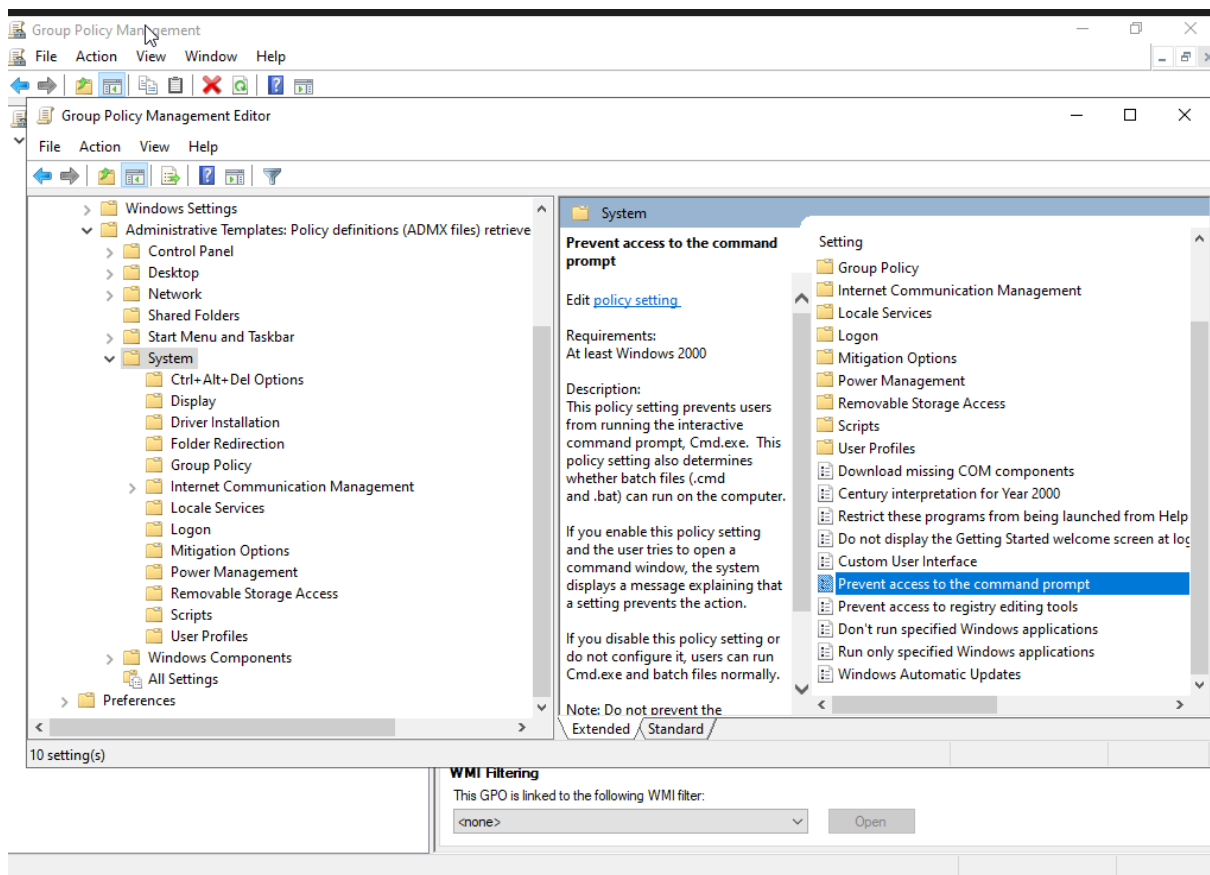


Fig 51: Navigate to the Group Policy Objects and verify the policies have been added to the policy.

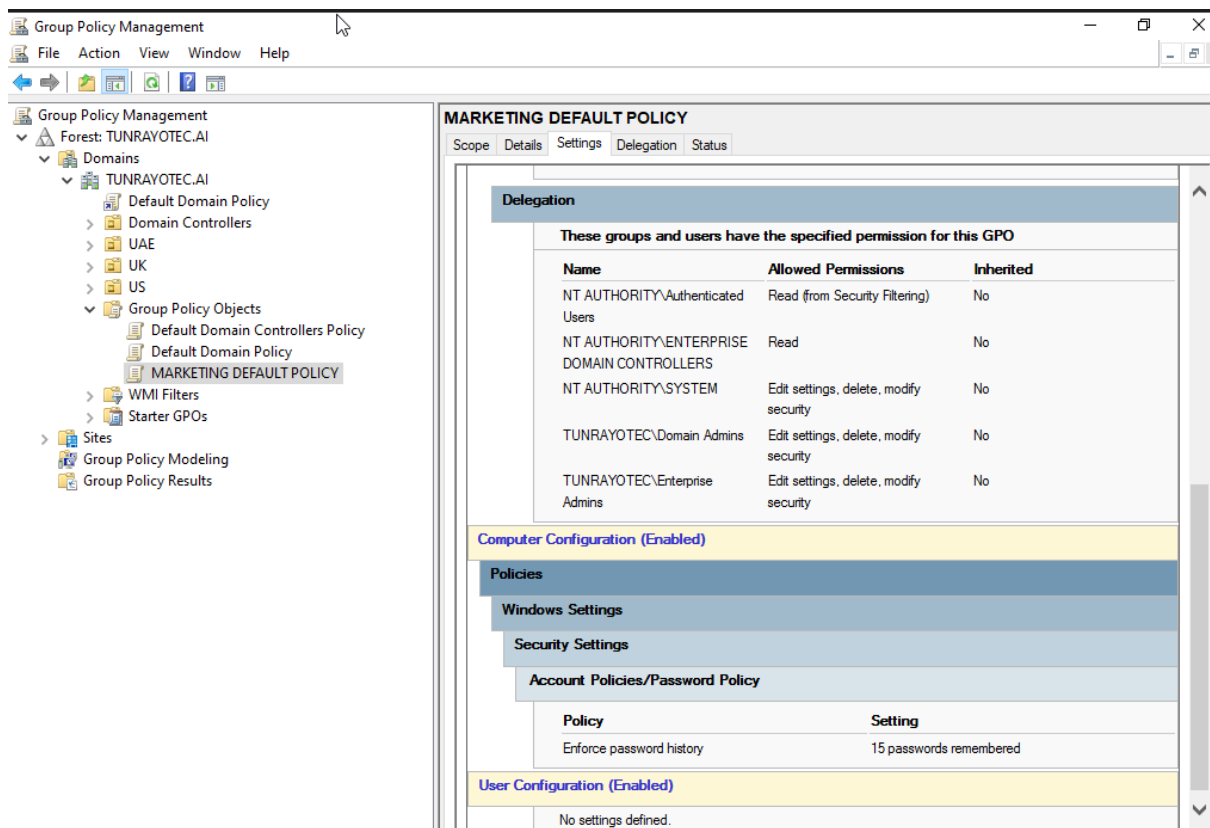


Fig 52: To link the Group Policy Objects to an individual, group, or domain. Navigate to the Forest, right click > link to an existing GPO>

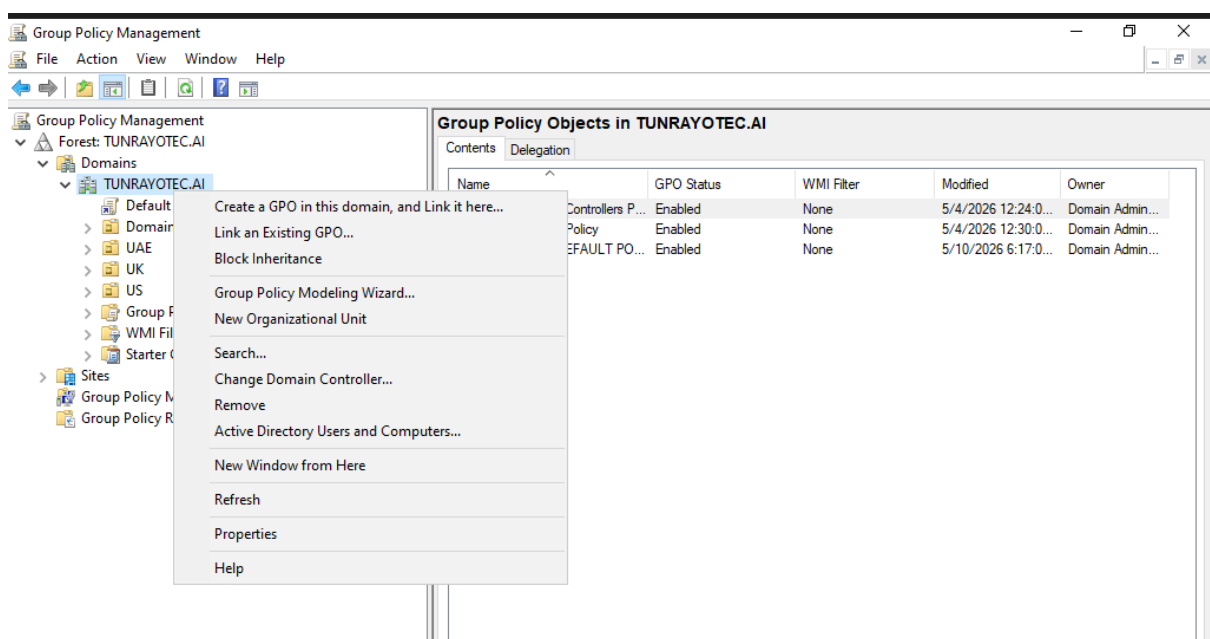


Fig 53: Link the policy to the Marketing Default Policy

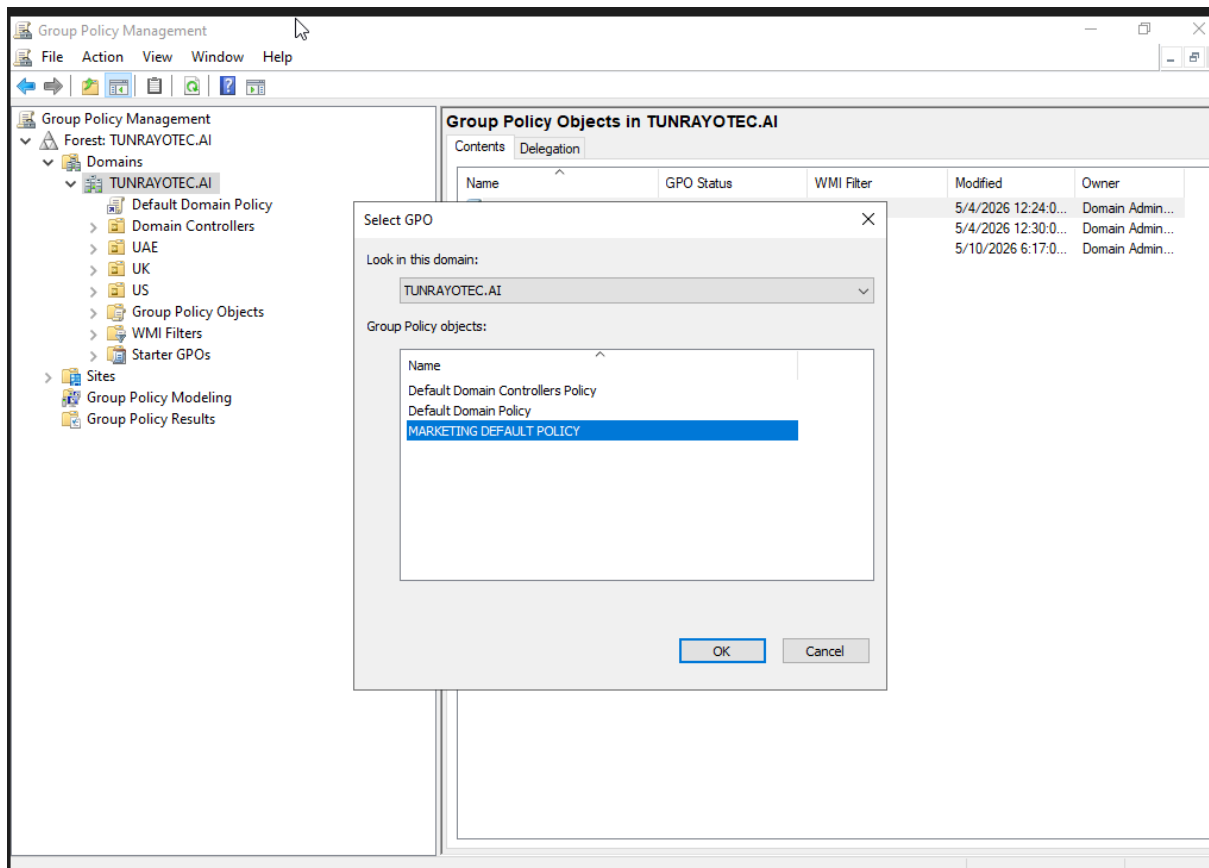


Fig 54: Navigate to the Marketing Default Policy >Security Filtering>Add>

Note: You can add the policy to either a group, groups, or an individual. If it is added to a group, the policy will be added to every member of that group. You can add it to an individual in a group whereby it does not affect any other user.

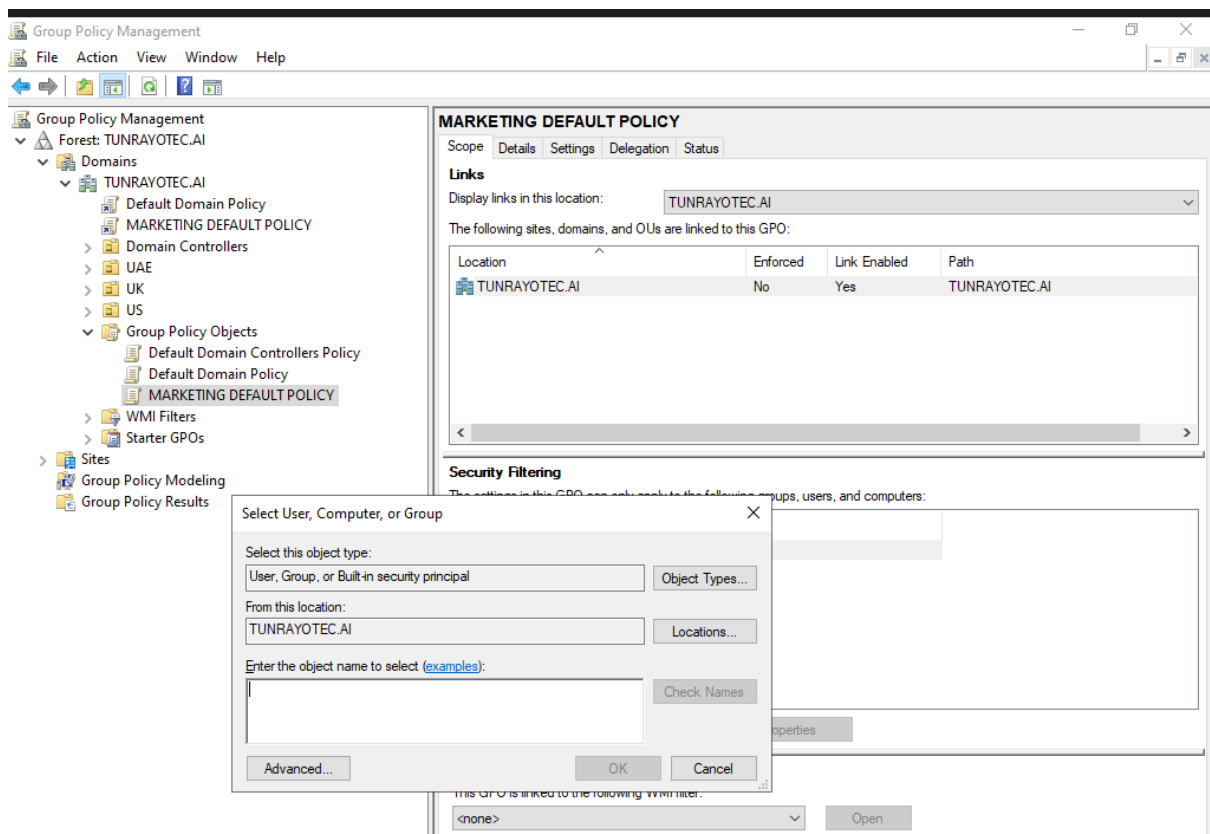


Fig 55: Navigate to Advanced> Find Now>Search for MARKETING>Click on Marketing>click OK.

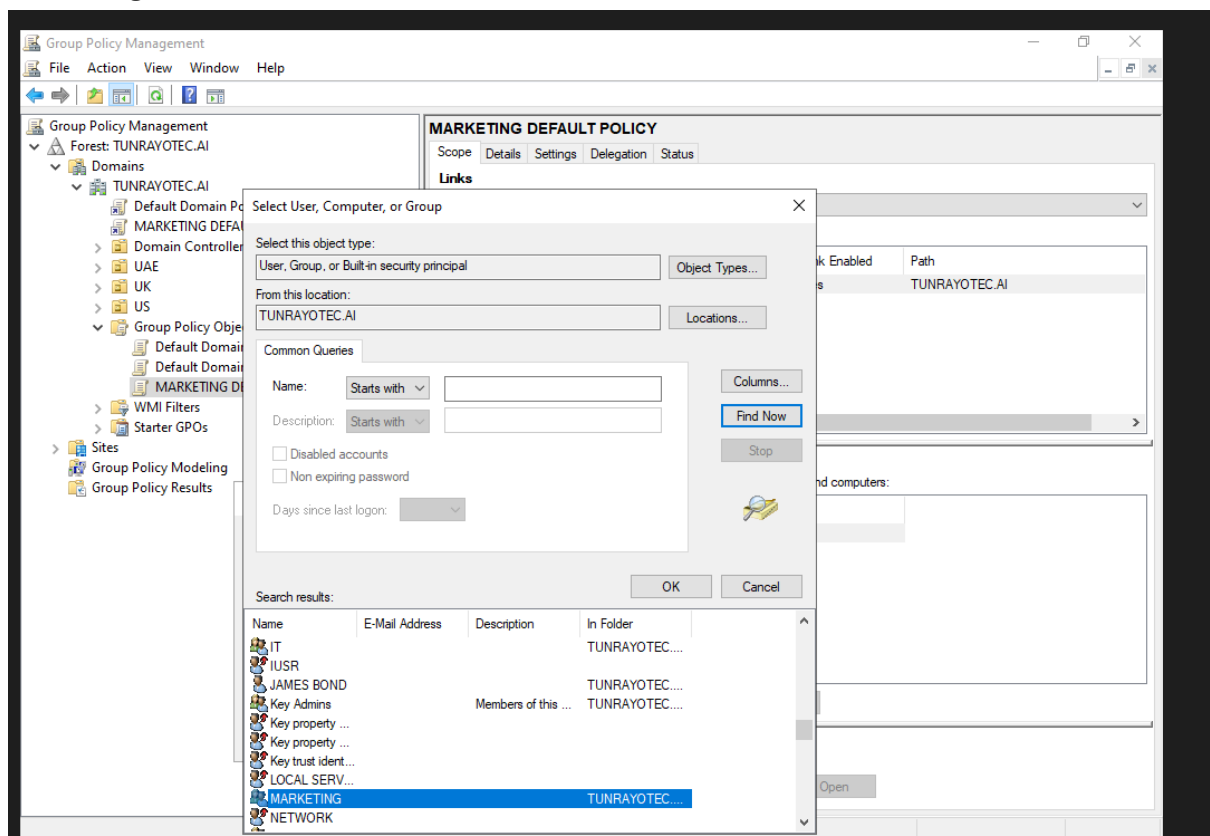


Fig 56:

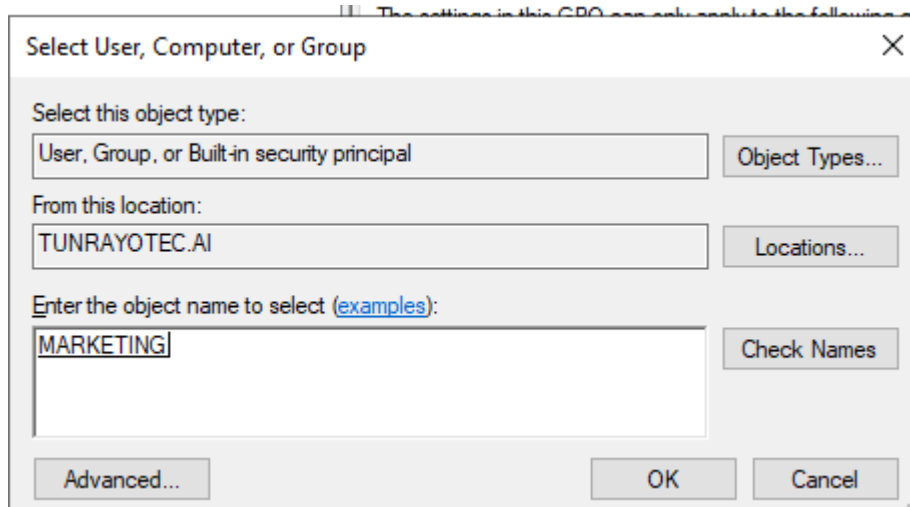


Fig 57: Navigate to CMD >enter gpupdate / force.

Note: This is to update the policy

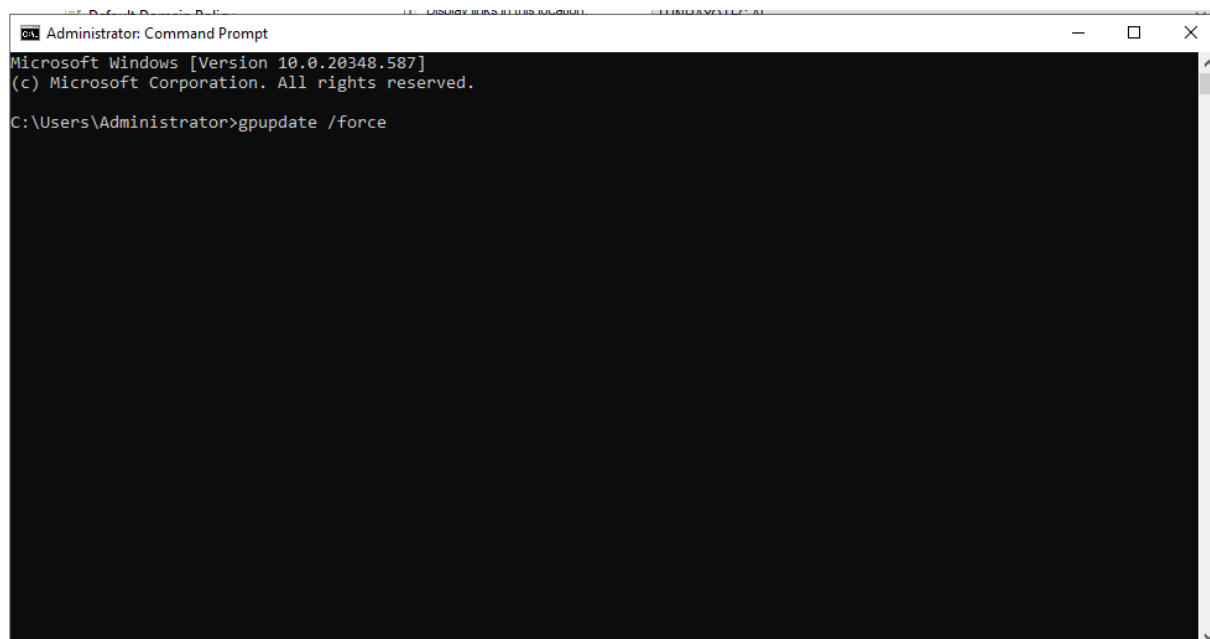
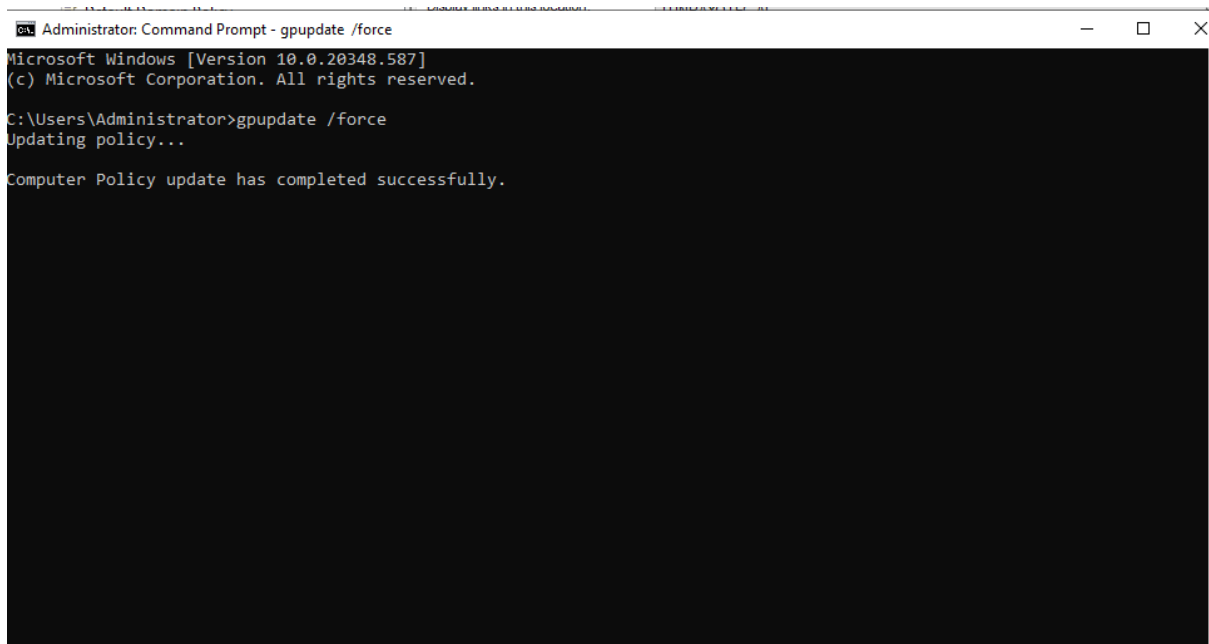


Fig 58:



```
Administrator: Command Prompt - gpupdate /force
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
```

Fig 59: Navigate to windows 8, navigate to CMD >enter gpupdate / force. Verify and validate the policies have been implemented.

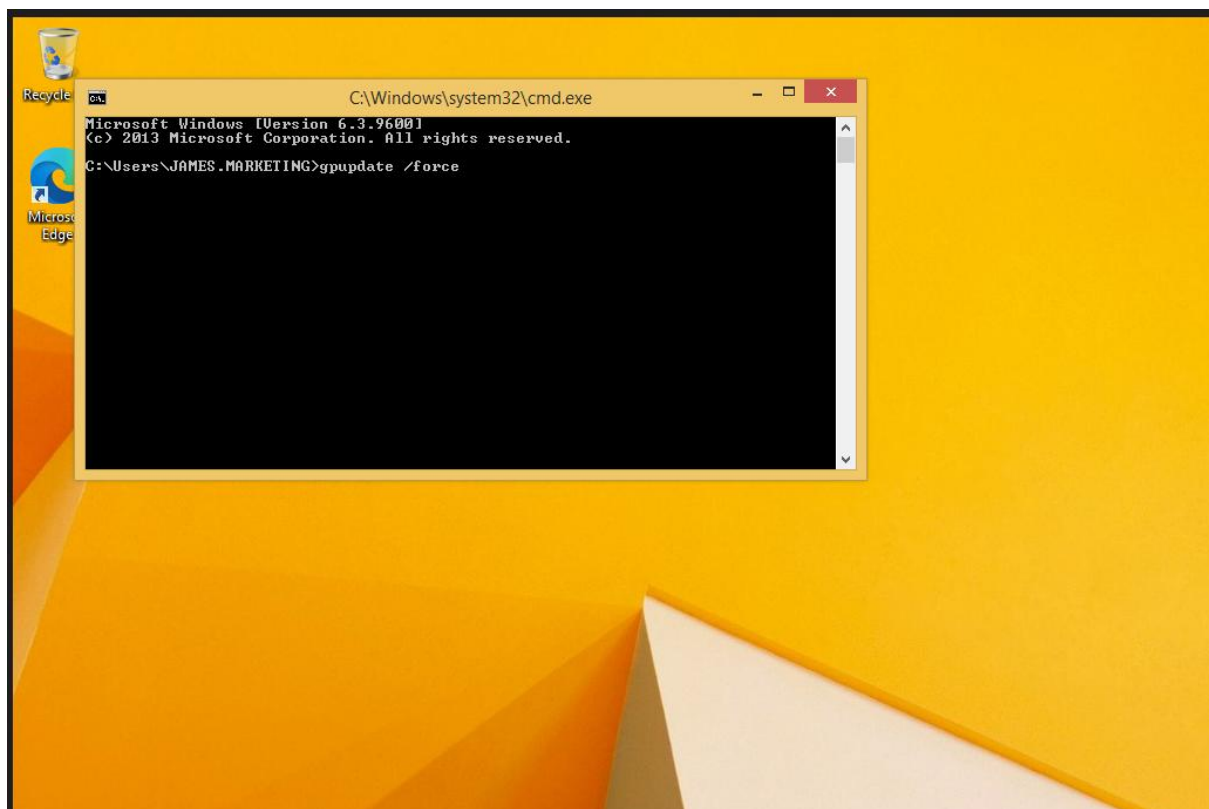
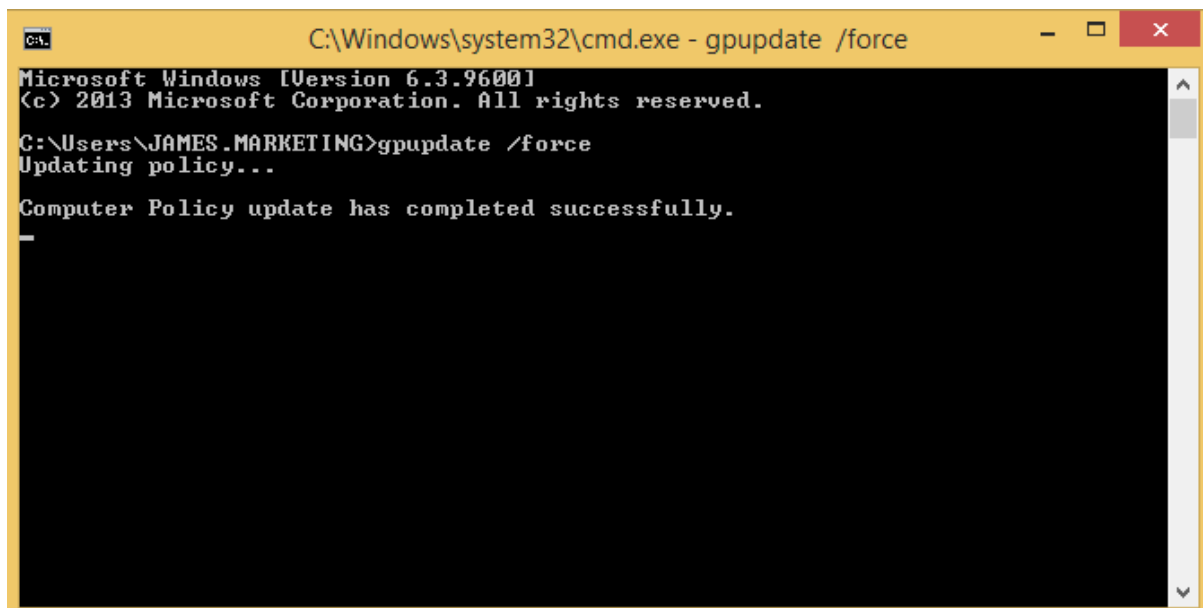


Fig 60:



```
C:\Windows\system32\cmd.exe - gpupdate /force
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\JAMES.MARKETING>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
_
```